



Pharmacy Chain Management System

Software Requirement Specification

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I. Record of Changes

Date	A* M, D	In charge	Change Description
24/05/2026	A	Team	Create initial SRS document structure
24/05/2026	A	Team	Add Context Diagram
31/05/2026	A	Team	Entity Relationship Diagram
31/5/2026	A	Team	Screen Flow Diagram
31/5/2026	A	HieuTT	Founder Feature
30/5/2026	A	LuongBD	Business Admin Feature
1/6/2026	A	HanND	Branch Manager Feature
1/6/2026	A	TruongPVD	Staff Feature
2/6/2026	A	HungNH	Inventory Manager Feature

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

The Pharmacy Chain Management System (PCMS) is a centralized software solution designed to support the operation and management of multiple pharmacy branches within a pharmacy chain. The system enables administrators, managers, inventory staff, and pharmacy staff to efficiently manage medicines, prescriptions, inventory, sales transactions, suppliers, and business performance.

The purpose of the system is to improve operational efficiency, reduce inventory shortages, enhance medicine traceability, and provide real-time business insights for management.

The Pharmacy Chain Management System serves as a unified platform that connects all pharmacy branches and stakeholders through a centralized database and management interface.

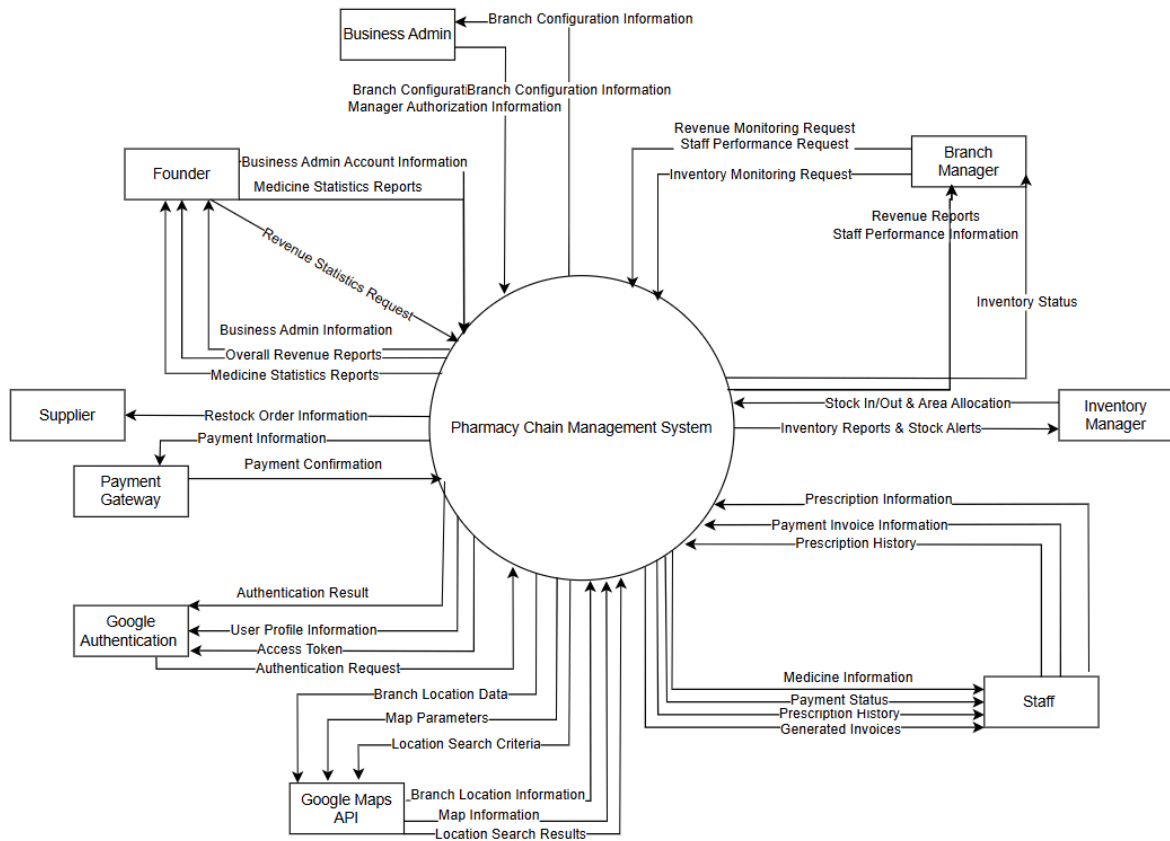


Figure 1-1: Pharmacy Chain Management System overview

1.2 System Scope

The system provides the following capabilities:

- User and role management
- Branch management
- Medicine management
- Inventory management
- Prescription management
- Invoice generation
- Revenue monitoring
- Supplier management
- Automated restocking
- Payment processing
- Reporting and analytics

The system supports multiple pharmacy branches and allows centralized monitoring of business operations across the entire pharmacy chain.

1.3 System Components

Founder Module

This module allows Founderistrators to:

- Manage user accounts
- Manage business administrator accounts
- Configure system settings
- Monitor overall system activities

Business Administration Module

This module allows Business Administrators to:

- Manage pharmacy branches
- Authorize branch managers
- Monitor overall business performance
- Review revenue and medicine statistics

Branch Management Module

This module allows Branch Managers to:

- View Branch Dashboard
- View Branch Revenue Statistics
- Monitor Staff Performance
- Monitor Branch Inventory

Inventory Management Module

This module allows Inventory Managers to:

- Receive Goods from Vendor
- QC Inspection
- Issue Stock to Stores
- Approve Internal Transfers
- Perform Stocktake
- Set Safety Stock Levels
- Identify Expired/Damaged Stock
- Track Batch & Lot Number
- View Inventory Reports

Prescription and Sales Module

This module allows Staff members to:

- Create prescriptions
- Search medicine information
- Generate invoices
- View prescription history
- Process customer transactions

Supplier Integration Module

This module allows the system to:

- Send automated restocking orders
- Receive supply confirmations
- Manage supplier information

Payment Processing Module

This module integrates with external payment services to:

- Process electronic payments
- Verify payment status
- Record transaction confirmations

1.4 External Entities

The Pharmacy Chain Management System interacts with the following external entities:

- Supplier
- Payment Gateway
- Google Maps API
- Google Authentication

2. User Requirements

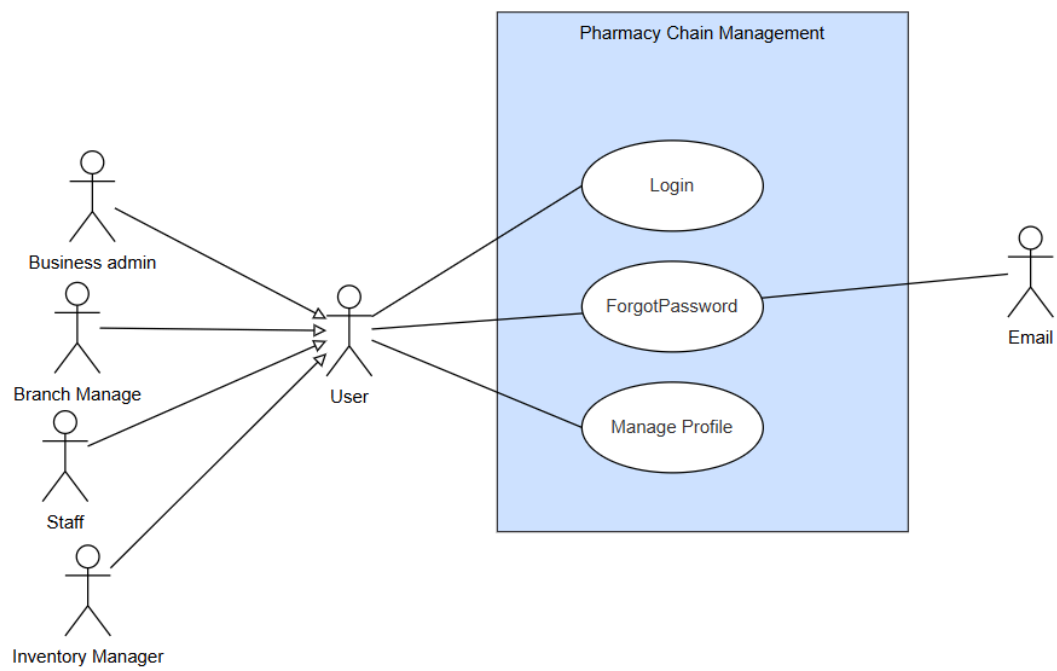
2.1 Actors

#	Actor	Description
1	Founder	Responsible for managing system configuration, user accounts, permissions, and system security.
2	Business Admin	Responsible for managing pharmacy branches, authorizing branch managers, and monitoring overall business performance.
3	Branch Manager	Responsible for supervising branch operations, monitoring branch revenue, monitoring branch inventory and evaluating staff performance.
4	Inventory Manager	Responsible for inventory control, stock allocation, stock monitoring, and inventory reporting.
5	Staff	Responsible for creating prescriptions, generating invoices, processing sales transactions, and viewing prescription history.
6	Payment Gateway	External payment service that processes online and electronic payments.
7	Google Maps API	External service responsible for providing map visualization, location search, route navigation, and branch location information.
8	Google Authentication	External service responsible for authenticating users through Google accounts and supporting secure login functionality.

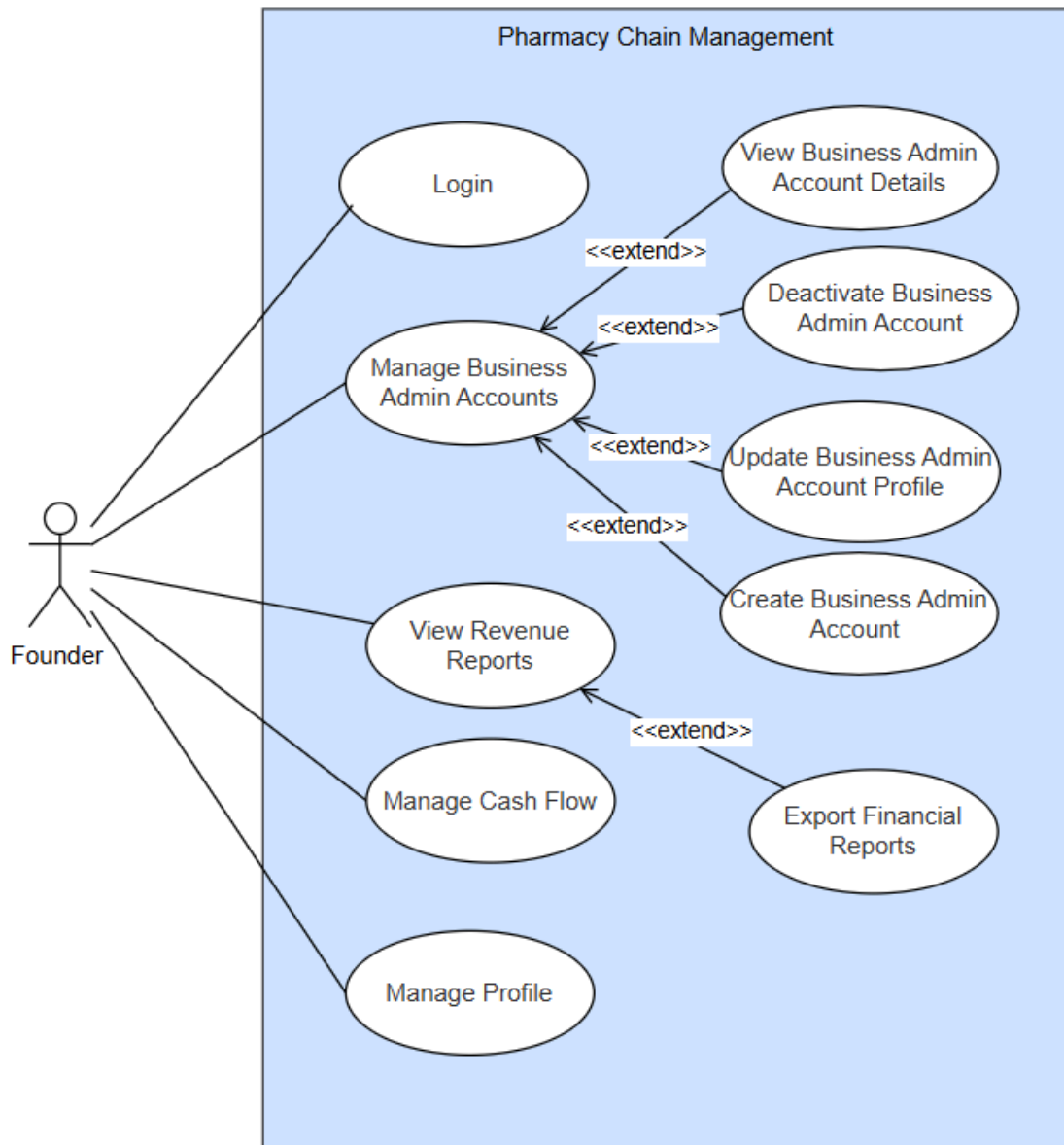
2.2 Use Cases

2.2.1 Diagram(s)

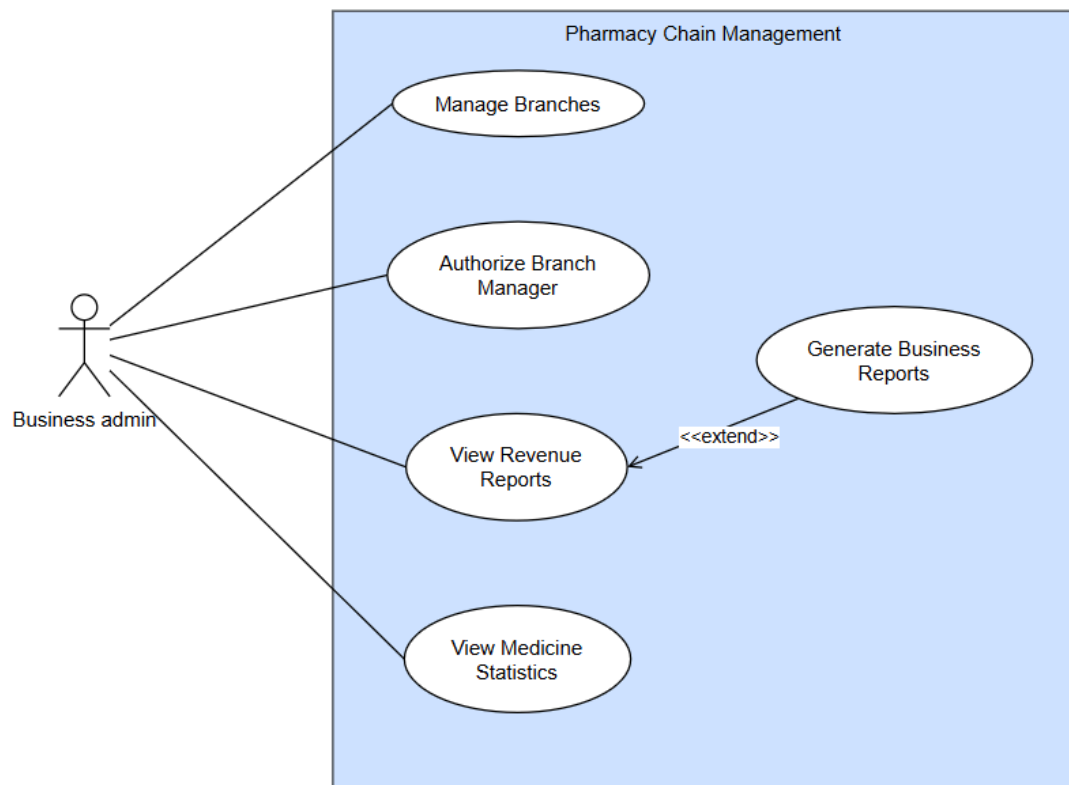
2.2.1.1 Use case of Common



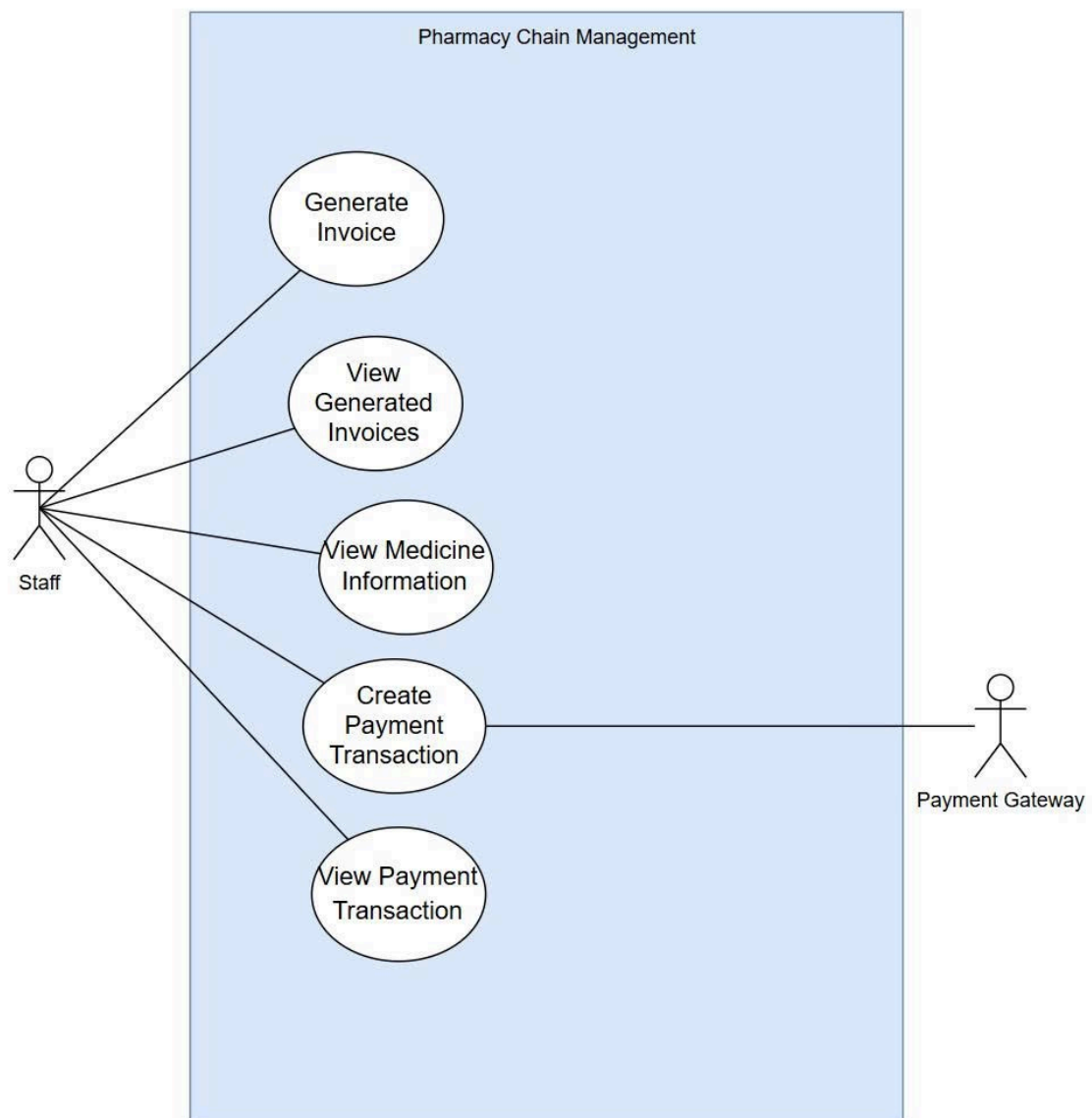
2.2.1.2 Use case of Founder



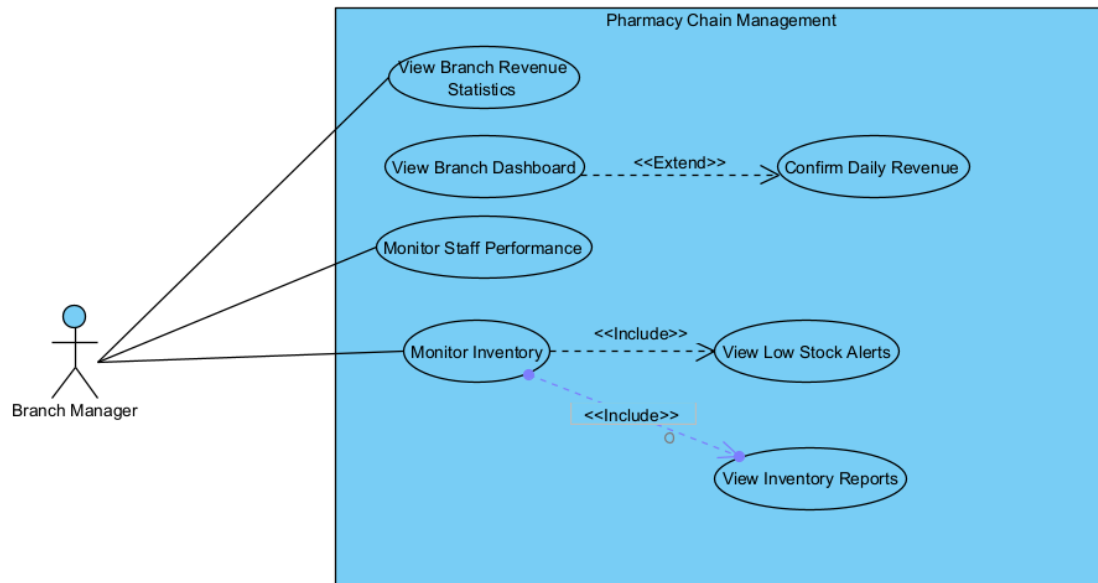
2.2.1.3 Use case of Business Admin



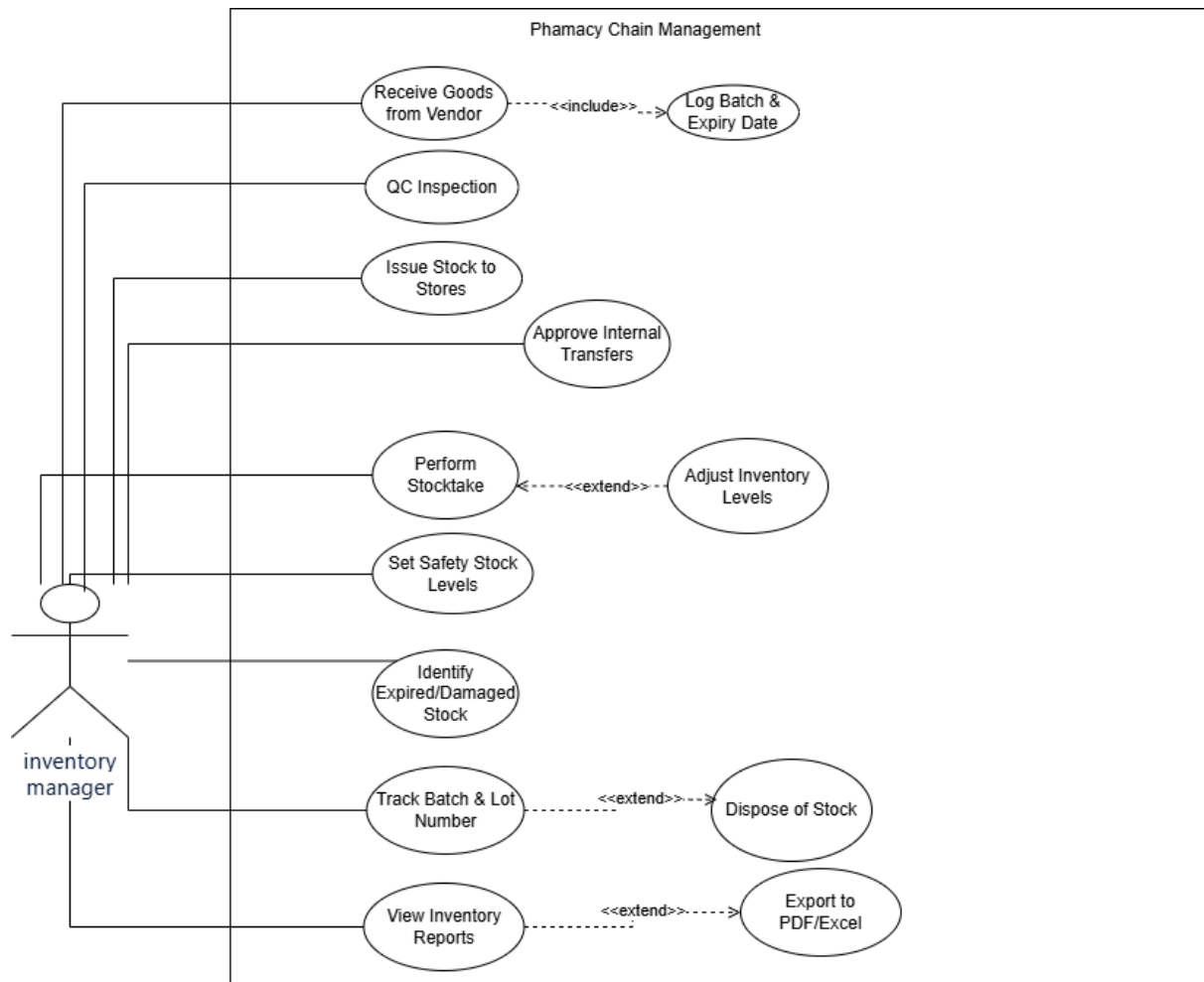
2.2.1.4 Use case of Staff



2.2.1.5 Use case of Branch Manager



2.2.1.6 Inventory Management



2.2.2 Descriptions

ID	Group function	Use Case	Actors	Use Case Description & Main Flow
UC01	Authentication	Login	All Internal Users	Authenticate user access to the system
UC02	Staff Management Account	Create Business Admin Account	Founder	Founder creates a new Business Admin account to grant management access to the system.
UC03	Branch Management	Manage Branches	Business Admin	Create and maintain pharmacy branch information
UC04	Branch Management	Authorize Branch Manager	Business Admin	Assign and authorize branch managers
UC05	Revenue Management	View Revenue Reports	Business Admin	Monitor overall revenue across branches
UC06	Analytics	View Medicine Statistics	Business Admin	Review medicine sales and inventory statistics
UC07	Inventory Monitoring	Receive Goods from Vendor	Inventory Manager	Record medicine arrivals from suppliers.
UC08	Inventory Monitoring	QC Inspection	Inventory Manager	Inspect the quality of received medicines before official entry.
UC09	DashBoard	View Branch DashBoard	Branch Manager	Access and monitor the branch dashboard to view key operational indicators, including revenue, staff performance, inventory alerts, branch efficiency metrics and confirm daily venue.
UC10	Reporting	View Branch Revenue Statistics	Branch Manager	View branch sales performance, revenue trends, and financial statistics for a selected reporting period.
UC11	Inventory Management	Issue Stock to Stores	Inventory Manager	Record medicine distribution and usage to pharmacy stores.
UC12	Inventory Management	Approve Internal Transfers	Inventory Manager	Transfer inventory and medicine between branches.
UC13	Inventory Management	View Inventory Reports	Inventory Manager	View and access inventory reports.
UC14	Inventory Management	Perform Stocktake	Inventory Manager	Periodically count physical stock against system records.
UC15	Sales Management	Generate Invoice	Staff	Create sales invoices for customers
UC16	Medicine Information	View Medicine Information	Staff	Search and view medicine information.
UC17	Payment Management	Create Payment Transaction	Staff, Payment Gateway	Create customer payments transaction

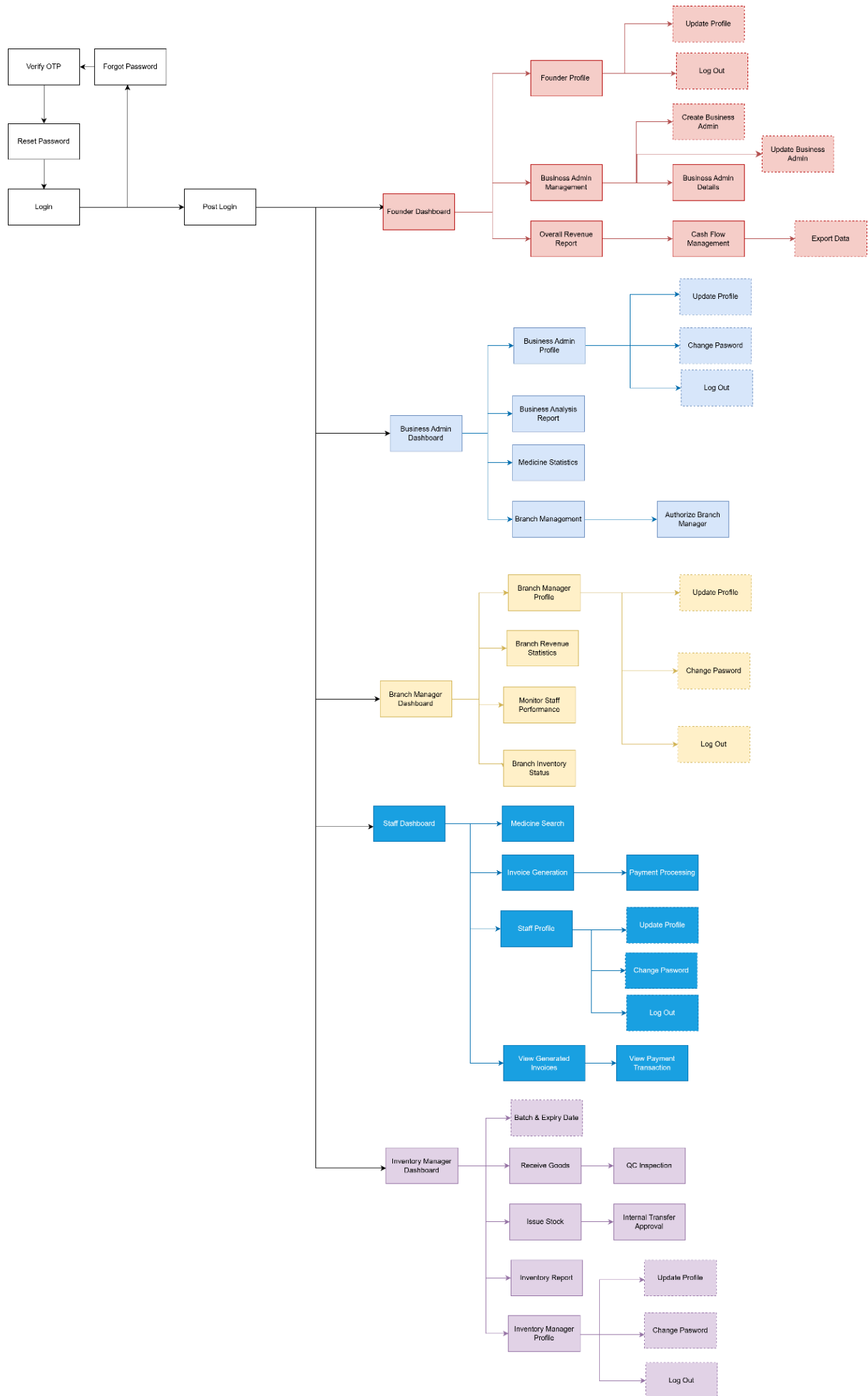
UC18	Sales Management	View Generated Invoices	Staff	Views previously generated invoices.
UC19	Inventory Management	Set Safety Stock Levels	Inventory Manager	Define minimum threshold levels to manage stock safety.
UC20	Supplier Management	Send Restock Orders	Supplier	Automatically send medicine replenishment orders
UC21	Supplier Management	Receive Supply Confirmation	Supplier	Confirm medicine delivery information
UC22	Reporting	Generate Business Reports	Business Admin	Generate business analysis and operational reports
UC23	Reporting	Revenue Reports	Founder	Access the system to view total revenue data and charts for the entire pharmacy chain, with options to filter by date or by specific branches.
UC24	Cash Flow Management	Manage Cash Flow	Founder	Track the system's cash flow statistics by calculating total cash inflows (sales) minus outflows (supplier payments, operating costs) to display actual profit or loss.
UC25	Reporting	Export Financial Reports	Founder	Select and export revenue and cash flow reports to Excel (.xlsx) or PDF formats for offline archiving.
UC26	Payment Management	View Payment Transaction	Staff	View customer payment transaction
UC27	Staff Management Account	Update Business Admin Account Profile	Founder	Founder updates the profile information of an existing Business Admin account to ensure data accuracy.
UC28	Staff Management Account	View Business Admin Account Details	Founder	Founder views the detailed profile and current status of a specific Business Admin account.
UC29	Staff Management Account	Deactivate Business Admin Account	Founder	Founder securely disables an existing Business Admin account, preventing any further access or operations by that user without deleting their historical data.
UC30	Inventory Management	Identify Expired/Damaged Stock	Inventory Manager	Detect and log medicines that are expired or damaged.
UC31	Inventory Management	Track Batch & Lot Number	Inventory Manager	Monitor and track batch and lot information of products.
UC32	Staff Monitoring	Monitor Staff Performance	Branch Manager	Monitor staff activities, productivity, and performance indicators within the assigned pharmacy branch

UC33	Inventory Monitoring	Monitor Branch Inventory	Branch Manager	Monitor branch inventory status, stock availability, and low-stock conditions to support branch operations.
UC34	Reporting	Business Analysis Report	Business Admin	View and analyze business performance data including revenue, sales trends, branch performance, and operational statistics across all pharmacy branches.
UC35	Authentication	Forgot Password	Business Admin, Branch Manager, Staff, Inventory Manager	Allows users to recover access to their account by verifying their identity through a registered email address, submitting a verification code, and establishing a new password.
UC36	Confirm Daily Revenue		Branch Manager	This use case allows the branch manager or accountant to confirm the daily revenue after checking the total sales recorded in the system. The user compares the system revenue with the actual received amount from cash, bank transfer, and other payment methods. If the amounts match, the revenue is confirmed. If there is a difference, the user must enter a reason before confirmation.

3. Software Feature

3.1 Functional Overview

3.1.1 Screens Flow



[Figure 3.1.1](#)

3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	Authentication	Login	Authenticate user access to the system.
2	Authentication	Forgot Password	Send password reset instructions via email.
3	User Profile	Profile	View personal account information and settings.
4	User Profile	Update Profile	Update personal profile information.
5	User Profile	Change Password	Change account password securely.
6	Dashboard	Founder Dashboard	Display overall system statistics, revenue, users, and branch information.
7	User Management	Business Admin Management	Manage Business Admin accounts.
8	User Management	Create Business Admin	Create a new Business Admin account.
9	User Management	Update Business Admin	Update Business Admin account information.
10	Reporting	Overall Revenue Report	View overall revenue reports of all pharmacy branches.

11	Finance	Cash Flow Management	Monitor income, expenses, profit, and cash flow.
12	Reporting	Export Financial Report	Export reports to Excel or PDF format.
13	Dashboard	Business Admin Dashboard	Display branch performance, medicine statistics, and business summaries.
14	Reporting	Business Analysis Report	Analyze business performance and operational statistics.
15	Analytics	Medicine Statistics	View medicine sales, inventory, and stock statistics.
16	Branch Management	Branch Management	Manage branch information and operations.
17	Branch Management	Authorize Branch Manager	Assign branch managers to pharmacy branches.
18	Dashboard	Branch Manager Dashboard	Display branch revenue, inventory status, and staff performance and daily revenue confirmation functions.
19	Reporting	Branch Revenue Statistics	View branch revenue and sales performance statistics.
20	Staff Management	Monitor Staff Performance	Monitor staff productivity and performance indicators.
21	Inventory	Branch Inventory Status	View medicine inventory status within a branch.
22	Dashboard	Staff Dashboard	Display daily sales and pharmacy operation summaries.

23	Medicine Management	Medicine Search	Search and view medicine information.
24	Sales Management	Invoice Generation	Create and manage customer invoices.
25	Payment Management	Payment Processing	Process customer payments through supported payment methods.
26	Sales Management	View Generated Invoices	View invoice history and invoice details.
27	Payment Management	View Payment Transactions	View payment transaction records and status.
28	Dashboard	Inventory Manager Dashboard	Display inventory overview and stock alerts.
29	Inventory Management	Receive Goods	Record medicines received from suppliers.
30	Inventory Management	Batch & Expiry Management	Manage medicine batch numbers and expiry dates.
31	Inventory Management	QC Inspection	Inspect medicine quality before inventory entry.
32	Inventory Management	Issue Stock	Distribute medicines to pharmacy branches.
33	Inventory Management	Internal Transfer Approval	Approve inventory transfers between branches.
34	Reporting	Inventory Report	View inventory reports and stock statistics.

3.1.3 Screen Authorization

#	Screen	Founder	Business Admin	Branch Manager	Inventory Manager	Staff
1	Login	x	x	x	x	x
2	Forgot Password		x	x	x	x
3	Profile	x	x	x	x	x
4	Update Profile	x	x	x	x	x
5	Change Password		x	x	x	x
6	Founder Dashboard	x				
7	Business Admin Management	x				
8	Create Business Admin	x				
9	Update Business Admin	x				
10	Overall Revenue Report	x				
11	Cash Flow Management	x				
12	Export Financial Report	x				
13	Business Admin Dashboard		x			
14	Business Analysis Report		x			
15	Medicine Statistics		x			

16	Branch Management		x			
17	Authorize Branch Manager		x			
18	Branch Manager Dashboard			x		
19	Branch Revenue Statistics			x		
20	Monitor Staff Performance			x		
21	Branch Inventory Status			x		
22	Staff Dashboard					x
23	Medicine Search					x
24	Invoice Generation					x
25	Payment Processing					x
26	View Generated Invoices					x
27	View Payment Transactions					x
28	Inventory Manager Dashboard				x	
29	Receive Goods				x	
30	Batch & Expiry Management				x	
31	QC Inspection				x	
32	Issue Stock				x	
33	Internal Transfer Approval				x	

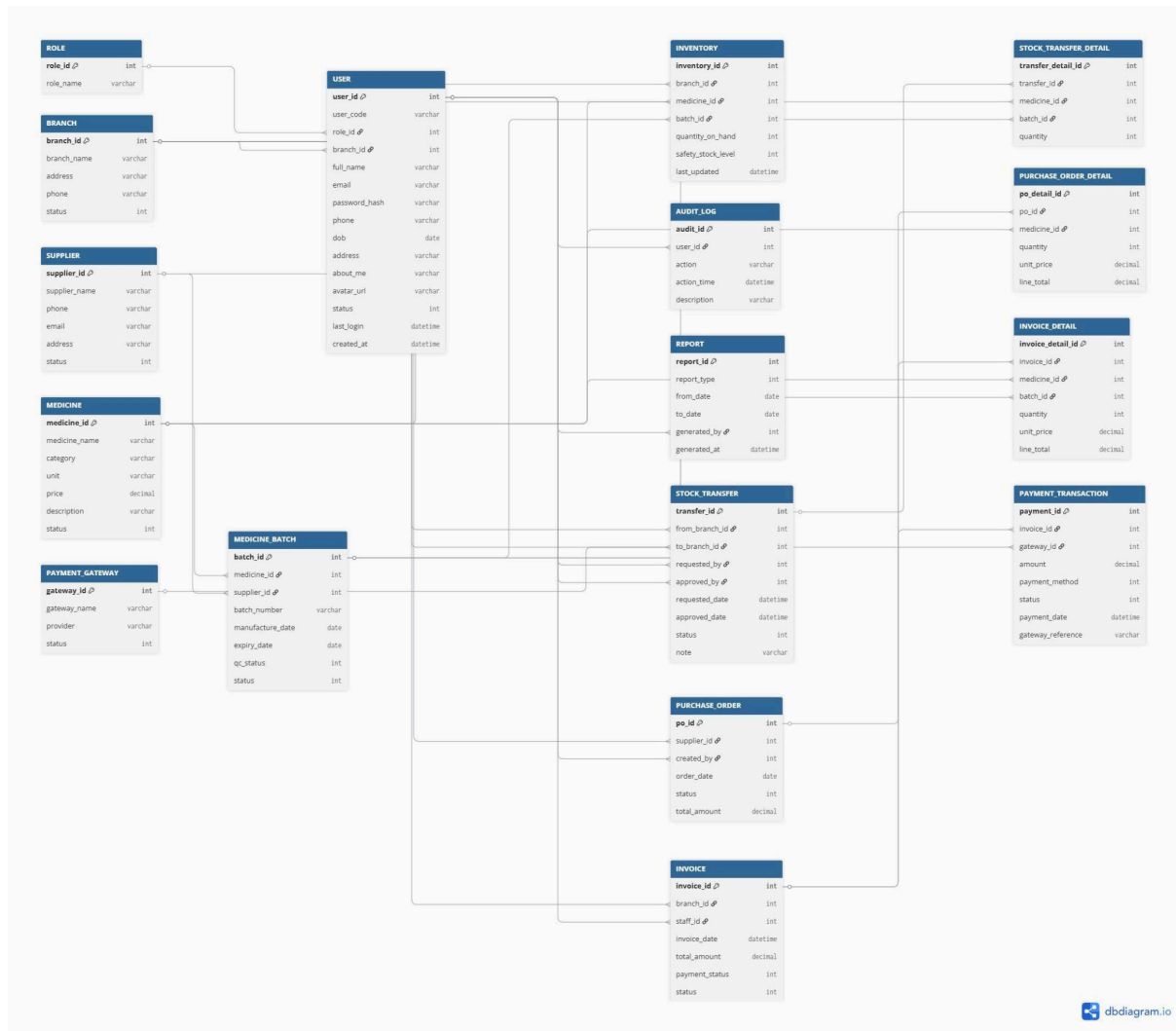
34	Inventory Report				x	
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3.1.4 Non-Screen Functions

#	Feature	System Function	Description
1	Authentication	User Authentication Service	Validates user credentials and manages user login sessions.
2	User Management	Role-Based Access Control Service	Manages access permissions based on user roles such as Founderistrator, Business Administrator, Branch Manager, Inventory Manager, and Staff.
3	Supplier Integration	Purchase Order Generation Service	Generates purchase orders for suppliers based on inventory requirements and stock levels.
4	Prescription Management	Prescription Validation Service	Validates prescription information before medicines are dispensed.
5	Sales Management	Invoice Generation Service	Automatically generates invoices and transaction records.
6	Sales Management	Revenue Calculation Service	Calculates sales revenue and key business performance metrics.

7	Payment Processing	Payment Gateway Integration Service	Integrates with external payment providers to process electronic payments.
8	Payment Processing	Payment Verification Service	Verifies payment completion status with external payment service providers.
9	Payment Processing	Payment Transaction Recording Service	Records payment transactions and maintains financial transaction logs.
10	Notification	Email Notification Service	Sends system-generated email notifications and alerts to users.
11	Audit & Compliance	Activity Monitoring Service	Monitors system-wide activities for operational oversight, security monitoring, and compliance purposes.

3.1.5 Entity Relationship Diagram



Entities Description

#	Entity	Description
1	User	Stores internal user account information for login, authentication, branch assignment, and role-based access control.
2	Role	Stores user role codes used for system authorization, such as Founder, Business Admin, Branch Manager, Inventory Manager, and Staff.
3	Supplier	Stores supplier information used for purchasing and restocking medicines.
4	Medicine	Stores medicine information such as medicine name, category, unit, price, description, and status.
5	Branch	Stores pharmacy branch information, including branch name, address, phone number, and operating status.

6	Medicine Batch	Stores medicine batch information, including batch number, supplier, manufacture date, expiry date, QC status, and batch status.
7	Inventory	Stores stock quantity of medicine batches available at each pharmacy branch.
8	Purchase Order	Records medicine purchase/restocking orders created by users and sent to suppliers.
9	Purchase Order Detail	Stores detailed medicine items, quantity, unit price, and line total for each purchase order.
10	Stock Transfer	Records stock transfer requests between pharmacy branches, including source branch, destination branch, requester, approver, and status.
11	Stock Transfer Detail	Stores detailed medicine batch items and quantities included in each stock transfer.
12	Invoice	Records sales invoices generated by staff at a pharmacy branch.
13	Invoice Detail	Stores medicine items, batch information, quantity, unit price, and line total for each invoice.
14	Payment Transaction	Records customer payment transactions related to invoices, including amount, method, status, payment date, and gateway reference.
15	Payment Gateway	Stores external payment gateway information used to process electronic payments.
16	Report	Stores generated report information, including report type, date range, creator, and generated time.
17	Audit Log	Records user activities in the system for monitoring, security, and audit purposes.

3.1.6 Entity Details

1. ROLE

#	Attribute name	PK	Type	Mandatory	Description
1	role_id	x	int	Yes	Unique identifier for the role.
2	role_name		string	Yes	Name of the role (e.g., Founder, Staff).

2. USER

#	Attribute name	P K	Type	Mandatory	Description
1	user_id	x	int	Yes	Unique user identifier.

2	role_id		int	Yes	Foreign key to ROLE.
3	branch_id		int	No	Foreign key to BRANCH (Null for Founder).
4	full_name		string	Yes	Full name of the user.
5	email		string	Yes	Login email address.
6	password_hash		string	Yes	Encrypted password.
7	phone		string	No	Contact number.
8	status		int	Yes	Active/Inactive flag.
9	created_at		datetime	Yes	Timestamp of account creation.

3. MEDICINE

#	Attribute name	PK	Type	Mandatory	Description
1	medicine_id	x	int	Yes	Unique medicine identifier.
2	medicine_name		string	Yes	Name of the medicine.
3	category		string	Yes	Medicine category (e.g., Antibiotic).
4	unit		string	Yes	Unit of measurement (e.g., Box, Pill).
5	price		decimal	Yes	Standard selling price.
6	description		string	No	Usage instructions or details.
7	status		int	Yes	Active/Discontinued flag.

4. MEDICINE_BATCH

#	Attribute name	PK	Type	Mandatory	Description
1	batch_id	x	int	Yes	Unique batch identifier.
2	medicine_id		int	Yes	Foreign key to MEDICINE.
3	supplier_id		int	Yes	Foreign key to SUPPLIER.
4	batch_number		string	Yes	Manufacturer's batch code.
5	manufacture_date		date	Yes	Date of manufacture.
6	expiry_date		date	Yes	Expiration date for FEFO tracking.
7	qc_status		int	Yes	Quality control status.
8	status		int	Yes	Batch availability status.

5. INVENTORY

#	Attribute name	PK	Type	Mandatory	Description
1	inventory_id	x	int	Yes	Unique inventory record ID.
2	branch_id		int	Yes	Foreign key to BRANCH.
3	medicine_id		int	Yes	Foreign key to MEDICINE.
4	batch_id		int	Yes	Foreign key to MEDICINE_BATCH.
5	quantity_on_hand		int	Yes	Current physical quantity.
6	safety_stock_level		int	Yes	Threshold for restocking alerts.
7	last_updated		datetime	Yes	Timestamp of last stock change.

6. STOCK_TRANSFER

#	Attribute name	PK	Type	Mandatory	Description
1	transfer_id	x	int	Yes	Unique transfer identifier.
2	from_branch_id		int	Yes	Source branch ID.
3	to_branch_id		int	Yes	Destination branch ID.
4	requested_by		int	Yes	User ID requesting the transfer.
5	approved_by		int	No	User ID approving the transfer.
6	requested_date		datetime	Yes	Date of request.
7	approved_date		datetime	No	Date of approval.
8	status		int	Yes	Pending/Approved/Completed.
9	note		string	No	Additional remarks.

7. STOCK_TRANSFER_DETAIL

#	Attribute name	PK	Type	Mandatory	Description
1	transfer_detail_id	x	int	Yes	Unique detail identifier.
2	transfer_id		int	Yes	Foreign key to STOCK_TRANSFER.
3	medicine_id		int	Yes	Foreign key to MEDICINE.
4	batch_id		int	Yes	Foreign key to MEDICINE_BATCH.
5	quantity		int	Yes	Amount of medicine transferred.

8. AUDIT_LOG

#	Attribute name	PK	Type	Mandatory	Description
1	audit_id	x	int	Yes	Unique audit log identifier.
2	user_id		int	Yes	Foreign key to USER who performed the action.
3	action		varchar	Yes	Type of action performed in the system.
4	action_time		datetime	Yes	Date and time when the action occurred.
5	description		varchar	No	Additional details about the action.

9. REPORT

#	Attribute name	PK	Type	Mandatory	Description
1	report_id	x	int	Yes	Unique report identifier.
2	report_type		int	Yes	Type/category of generated report.
3	from_date		date	Yes	Start date of the reporting period.
4	to_date		date	Yes	End date of the reporting period.
5	generated_by		int	Yes	Amount of medicine transferred.
6	generated_at		datetime	Yes	Date and time when the report was generated.

10. INVOICE

#	Attribute name	PK	Type	Mandatory	Description
1	invoice_id	x	int	Yes	Unique invoice identifier.

2	branch_id		int	Yes	Foreign key to BRANCH where the sale occurred.
3	staff_id		int	Yes	Foreign key to USER (Staff) who created the invoice.
4	invoice_date		datetime	Yes	Date and time when the invoice was created.
5	total_amount		decimal	Yes	Total amount of the invoice.
6	payment_status		int	Yes	Payment status of the invoice.
7	status		int	Yes	Current invoice status.

11. INVOICE_DETAIL

#	Attribute name	PK	Type	Mandatory	Description
1	invoice_detail_id	x	int	Yes	Unique invoice detail identifier.
2	invoice_id		int	Yes	Foreign key to INVOICE.
3	medicine_id		int	Yes	Foreign key to MEDICINE.
4	batch_id		int	Yes	Foreign key to MEDICINE_BATCH.
5	quantity		int	Yes	Amount of medicine transferred.
6	unit_price		decimal	Yes	Selling price per unit.
7	line_total		decimal	Yes	Total amount for the invoice item.

12. PAYMENT_TRANSACTION

#	Attribute name	PK	Type	Mandatory	Description
---	----------------	----	------	-----------	-------------

1	payment_id	x	int	Yes	Unique payment transaction identifier.
2	invoice_id		int	Yes	Foreign key to INVOICE.
3	gateway_id		int	Yes	Foreign key to PAYMENT_GATEWAY.
4	amount		decimal	Yes	Foreign key to MEDICINE_BATCH.
5	payment_method		varchar	Yes	Method used for payment.
6	status		int	Yes	Current transaction status.
7	payment_date		datetime	Yes	Date and time of payment.
8	gateway_reference		varchar	No	Reference code returned by the payment gateway.

13. PAYMENT_GATEWAY

#	Attribute name	PK	Type	Mandatory	Description
1	gateway_id	x	int	Yes	Unique payment gateway identifier.
2	gateway_name		varchar	Yes	Name of the payment gateway.
3	provider		int	Yes	Foreign key to MEDICINE.
4	status		int	Yes	Foreign key to Current gateway status (active/inactive).

3.2. Authentication Management

3.2.1 <<UC01- Login >>

3.2.1.1 Screen Mock-up

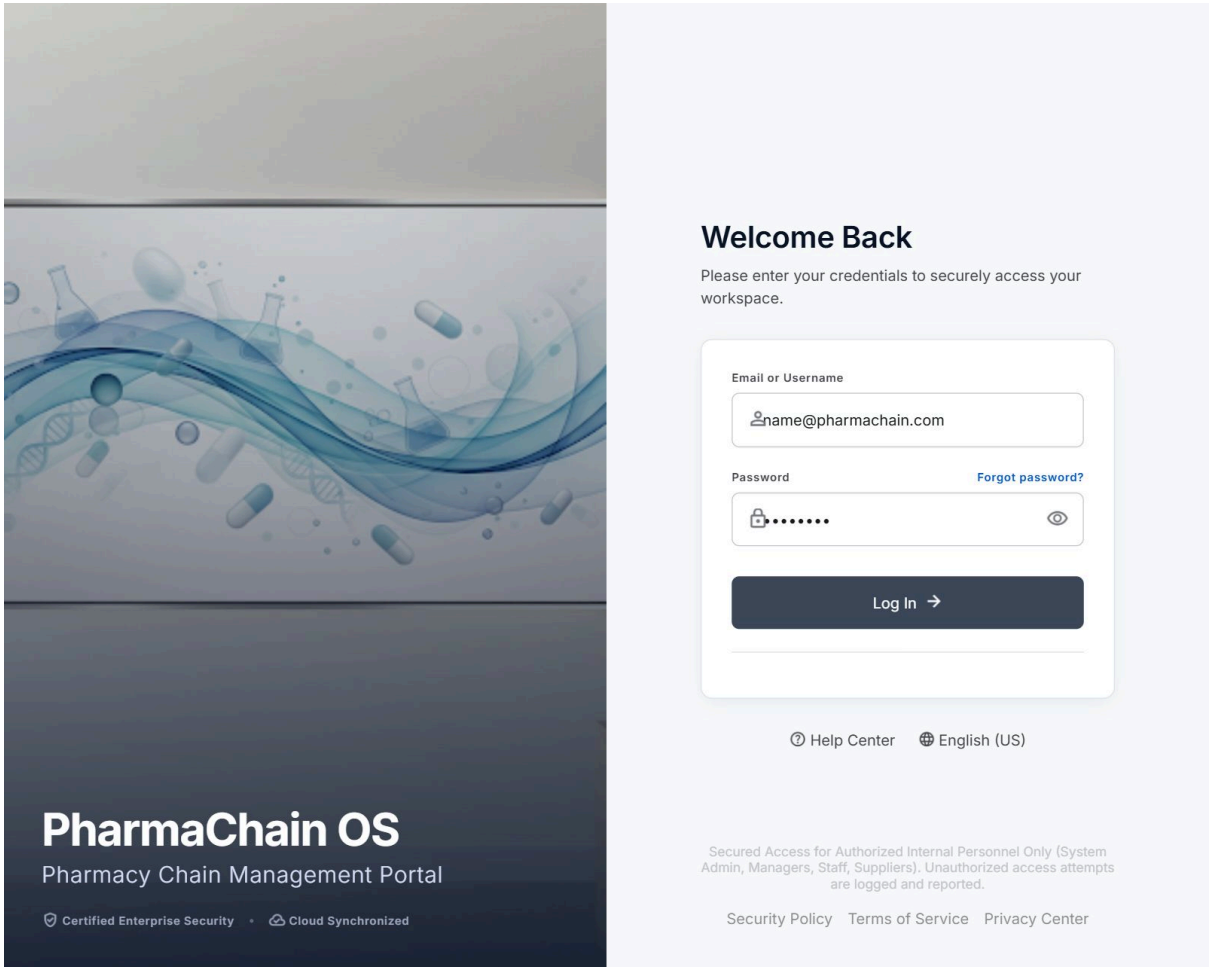


Table 3-2-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Username	Text	Yes	50	Unique identifier for the internal user account.
2	Password	Text (Masked)	Yes	32	Secure credential matching the provided Username.
3	Role Authorization	Drop-down	Yes	-	Used to route validation logic to internal config or central repository (BR-01, BR-02).
4	Remember Me	Checkbox	No	-	Optional flag to persist the active session state.

5	Login	Button	-	-	Triggers the authentication validation process.
6	Forgot Password	Button / Link	-	-	Redirects users to the password recovery flow.

3.2.1.2 Use Case Description

Use Case ID	UC01	Use Case Name		Authenticate User	
Author	Trần Trung Hiếu	Version	1.0	Date	03/06/2026
Actor	Primary Actor: Internal User (Founder, Business Admin, Branch Manager, Inventory Manager, Staff)				
Description	The system authenticates an internal user to grant secure access to their respective modules based on their assigned role.				
Precondition	PRE-01: The Internal User is not currently authenticated and the system is ready to accept login requests.				
Trigger	TRG-01: The Internal User requests access to the secured areas of the system.				
Post-Condition	POS-01: The Internal User is successfully authenticated, a secure session is established, and access is granted.				
Main flows					
Step	Actor		Action		
1	User		Requests to authenticate into the system.		
2	System		Prompts for the necessary authentication credentials (e.g., Username and Password).		
3	User		Provides the requested authentication credentials.		
4	System		Validates the provided credentials based on the user's role constraints (BR-01, BR-02, BR-03).		

5	System	Records the successful authentication, establishes an active session, and grants access to the system.
---	--------	--

Alternative flows

4a	Invalid Credentials (Violates BR-01 or BR-02)	
Sub step	Actor	Action
4a.1	System	Notifies the Internal User of the authentication failure and prompts to re-provide the credentials.
4a.2	User	Provides the credentials again (Returns to step 4).
4b	Account Inactive or Locked (Violates BR-03)	
Sub step	Actor	Action
4b.1	System	Notifies the Internal User that the account is inactive or locked and provides instructions to contact support.
4b.2	System	Terminates the authentication process (Use case ends).
*a	Cancellation (Can occur at any step before step 5)	
Sub step	Actor	Action
*a.1	User	Requests to cancel the authentication process.
*a.2	System	Terminates the process and returns to the initial public state.

Business Rules

#	Rule Description
BR-01	Founder Authentication: If the user is identified as a Founder, the system must validate the credentials against the secure internal system configuration logic (bypassing the standard database).

BR-02	Standard User Authentication: For all other roles (Business Admin, Branch Manager, Inventory Manager, Staff, Supplier), the system must validate the credentials against the central user repository.
BR-03	Account Status Validation: The target account must be in an "Active" state. Inactive, suspended, or locked accounts must be denied access regardless of credential correctness.

3.2.2 <<UC- Manage Profile >>

3.2.2.1 Screen Mock-up

Manage Profile

Nguyen Van A

Branch Manager

Home > My Profile > Manage Profile

Manage Profile

View and update your personal information.

Profile Information

Change Password

Account Security

Full Name *

Nguyen Van A

Email *

nguyenvana@pharmacy.com

Role

Branch Manager

Phone

0123 456 789

Branch *

FPT Pharmacy - Cau Giay

Date of Birth

15/08/1990

Address

123 Cau Giay Street, Cau Giay District, Hanoi

About Me

Experienced in pharmacy operations and inventory management.

Save Changes

Change Photo

JPG, PNG or GIF. Max size 2MB.

Account Information

User ID

U00012345

Status

Active

Last Login

27/05/2025 09:15 AM

Joined Date

10/02/2024

Quick Actions

Change Password

Account Security

Table 3-2-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description

1	Profile Photo	Upload	No	2MB	Upload user avatar image. Only JPG, PNG formats are accepted.
2	Full Name	Text	Yes	100	User's full name displayed and updated by the account owner.
3	Email	Text	Yes	-	User email address. Must follow valid email format and be unique within the system.
4	Role	Drop-down (Read-only)	Yes	-	Displays the role assigned to the current user (Founder, Business Admin, Branch Manager, Inventory Manager, Staff).
5	Phone Number	Text	-	-	User contact number. Must follow a valid phone number format.
6	Branch	Drop-down	No	-	Displays the assigned pharmacy branch. Editable only if permitted by role authorization.
7	Date of Birth	Date Picker	No	-	User's date of birth.
8	Address	Text Area	No	-	Residential address of the user.
9	About Me	Text Area	No	-	Additional personal description or profile summary.
10	User ID	Label (Read-only)	Yes	-	Unique identifier of the user account generated by the system.
11	Account Status	Status Label	Yes	-	Displays current account status (Active, Inactive, Locked).

12	Last Login	Label (Read-only)	No	-	Displays the most recent login date and time.
13	Joined Date	Label (Read-only)	Yes	-	Displays the account creation date.
14	Change Photo	Button	-	-	Opens file selection dialog for uploading a new profile image.
15	Save Changes	Button	-	-	Validates profile information and saves updates to the system.
16	Change Password	Button	-	-	Redirects the user to the Change Password screen.
17	Account Security	Button	-	-	Redirects the user to Account Security settings (2FA, authentication settings, etc.).

3.2.2.2 Use Case Description

Use Case ID	UC04	Use Case Name		Manage Profile	
Author	Bùi Đức Lương, Phạm Vũ Đình Trường	Version	1.0	Date	03/06/2026
Actor	Primary Actor: Internal User (Founder, Business Admin, Branch Manager, Inventory Manager, Staff)				
Description	This use case allows authenticated users to view and update their personal profile information, including contact information, address, profile photo, and personal details.				
Precondition	Internal User is successfully authenticated and has an active session.				
Trigger	Internal User selects the "Manage Profile" option from the system menu. .				
Post-Condition	User's profile information is successfully updated and stored in the system database				
Main flows					

Step	Actor	Action
1	User	Selects the Manage Profile function
2	System	Displays the current user profile information
3	User	Reviews profile information.
4	User	Updates editable profile fields (Full Name, Phone Number, Address, Date of Birth, Profile Photo, etc.)..
5	User	Clicks the Save Changes button.
6	System	Validates the entered information .
7	System	Saves the updated profile information to the database.
8	System	Displays a success confirmation message.

Alternative flows

AF-01	Invalid Data Format	
Sub step	Actor	Action
1	System	Detects invalid input data (e.g., invalid email format or phone number).
2	System	Displays validation error message.
3	User	Corrects the information and resubmits the update request.
AF-02	Profile Photo Upload Failure	
Sub step	Actor	Action
1	User	Uploads a profile image
2	System	Detects unsupported file type or file size exceeding the limit.
3	System	Displays an error message and rejects the upload.
4	User	Selects another valid image file.
EX-01	Database Update Failure	
Sub step	Actor	Action
1	System	Database Update Failure
2	System	Displays an error message.
3	System	Maintains the original profile information without modification.

Business Rules

#	Rule Description
BR-01	Full Name is required and must not exceed 100 characters
BR-02	Email must be in a valid email format.
BR-03	Phone Number must contain valid numeric characters.
BR-04	Profile Photo must be in JPG or PNG format and must not exceed 2 MB.
BR-05	Role information cannot be modified through the Manage Profile screen.
BR-06	TheUser ID, Account Status, Last Login, and Joined Date are read-only fields.

3.2.3 <<UC02-Forgot Password >>

3.2.3.1 Screen Mock-up

Pharmacy Chain Management System [← Back to Login](#)

Forgot Password?

Don't worry! You can reset your password in a few simple steps.

How it works

- 1. Enter your email**
Provide the email address associated with your account.
- 2. Check your email**
We'll send you a verification code to your email.
- 3. Reset your password**
Enter the code and create a new password to regain access.

This feature is available for Business Admin, Branch Manager, Inventory Manager, and Staff accounts. System Admin accounts are managed directly by the system.

Step 1: Verify Your Email
Enter the email address associated with your account.

Email Address *

✉ Enter your email address

ℹ We will send a verification code to this email address.

Send Verification Code

or

[← Back to Login](#)

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Table 3-1: Screen Definition – Forgot Password (Verify Email)

#	Field Name	Type	Mandatory	Max Length	Description

1	Email Address	Text	Yes	100	Registered email address associated with the user account.
2	Send Verification Code	Button	-	-	Sends a verification code to the provided email address.
3	Back to Login	Button / Link	-	-	Redirects the user to the Login screen.
4	Progress Indicator	Label	-	-	Displays the current password recovery step.


← Back to Login

Forgot Password?

Don't worry! You can reset your password in a few simple steps.



How it works

- 1. Enter your email**
Provide the email address associated with your account.
- 2. Check your email**
We'll send you a verification code to your email.
- 3. Reset your password**
Enter the code and create a new password to regain access.

This feature is available for Business Admin, Branch Manager, Inventory Manager, and Staff accounts. System Admin accounts are managed directly by the system.

1

2

3

4

Verify Email

Verify Code

Reset Password

Complete



Step 2: Verify Code

We have sent a 6-digit verification code to **nguyenvana@pharmacy.com**

Verification Code *

Didn't receive the code? [Resend Code](#) (56s)

The verification code will expire in 10 minutes.

← Back
Verify Code

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Table 3-2: Screen Definition – Verify Code

#	Field Name	Type	Mandatory	Max Length	Description
1	Verification Code	OTP Input	Yes	6	Verification code received through email.

2	Verify Code	Button	-	-	Validates the entered verification code.
3	Resend Code	Button / Link	-	-	Requests a new verification code.
4	Countdown Timer	Label	-	-	Displays the remaining validity period of the verification code.
5	Back	Button	-	-	Returns to the previous screen.


Pharmacy Chain
Management System

[← Back to Login](#)

Forgot Password?

Don't worry! You can reset your password in a few simple steps.



How it works

- 1. Enter your email**
Provide the email address associated with your account.
- 2. Check your email**
We'll send you a verification code to your email.
- 3. Reset your password**
Enter the code and create a new password to regain access.

 This feature is available for Business Admin, Branch Manager, Inventory Manager, and Staff accounts.
System Admin accounts are managed directly by the system.

1

2

3

4

Verify Email
Verify Code
Reset Password
Complete



Step 3: Reset Your Password

Create a new password for your account.

New Password *



- At least 8 characters long
- Contains at least one uppercase letter
- Contains at least one number
- Contains at least one special character

Confirm New Password *



← Back

Reset Password

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Table 3-3: Screen Definition – Reset Password

#	Field Name	Type	Mandatory	Max Length	Description
1	New Password	Text (Masked)	Yes	32	New password that satisfies the system password policy.

2	Confirm New Password	Text (Masked)	Yes	32	Re-entered password for confirmation.
3	Show/Hide Password	Toggle Icon	No	-	Displays or hides password values.
4	Password Requirements	Label	-	-	Displays password policy requirements.
5	Reset Password	Button	-	-	Updates the user's password.
6	Back	Button	-	-	Returns to the Verify Code screen.


Pharmacy Chain
Management System

[← Back to Login](#)

Forgot Password?

Don't worry! You can reset your password in a few simple steps.



How it works

- 1. Enter your email**
Provide the email address associated with your account.
- 2. Check your email**
We'll send you a verification code to your email.
- 3. Reset your password**
Enter the code and create a new password to regain access.

This feature is available for Business Admin, Branch Manager, Inventory Manager, and Staff accounts. System Admin accounts are managed directly by the system.

✓

Verify Email

✓

Verify Code

✓

Reset Password

4

Complete



Password Reset Successful!

Your password has been reset successfully.
You can now log in to your account with your new password.

Go to Login

[Return to Home](#)

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Table 3-4: Screen Definition – Complete

#	Field Name	Type	Mandatory	Max Length	Description
1	Success Icon	Image/Icon	-	-	Indicates successful password reset operation.

2	Success Message	Label	-	-	Displays confirmation that the password has been reset successfully.
3	Go To Login	Button	-	-	Redirects the user to the Login screen.
4	Return To Home	Button	-	-	Redirects the user to the Home page.
5	Progress Indicator	Label	-	-	Displays completion status of the password recovery process.

3.2.3.2 Use Case Description

Use Case ID	UC35	Use Case Name		Forgot Password	
Author	Trần Trung Hiếu	Version	1.0	Date	03/06/2026
Actor	Primary Actor: Internal User (Business Admin, Branch Manager, Inventory Manager, Staff)				
Description	Allows users to recover access to their account by verifying their identity through a registered email address, submitting a verification code, and establishing a new password.				
Precondition	PRE-01: The user account exists in the system and is not currently authenticated.				
Trigger	TRG-01: The user requests to recover their account password.				
Post-Condition	POS-01: The user's password is successfully updated, securely recorded, and ready for future authentication.				
Main flows					
Step	Actor		Action		

1	User	Requests to recover the account password.
2	System	Prompts for the registered email address.
3	User	Provides the registered email address and requests verification.
4	System	Validates the email address (BR-01, BR-02) and dispatches a secure verification code.
5	System	Prompts the user for the verification code.
6	User	Provides the received verification code.
7	System	Validates the verification code constraints (BR-03, BR-04).
8	System	Prompts for a new password and password confirmation.
9	User	Provides and confirms the new password.
10	System	Validates the new password against security requirements (BR-05, BR-06, BR-07).
11	System	Updates the account password, records the activity (BR-08), and notifies the user of the successful password reset.
12	User	Acknowledges the completion.
13	System	Returns the user to the initial authentication state.

Alternative flows

AF-01	Invalid Email Address (Branches from Step 4)	
Sub step	Actor	Action
AF-01.1	System	Detects invalid email format, non-existing email address, or policy violation (BR-01, BR-02).

AF-01.2	System	Notifies the user of the error and prompts for correction.
AF-01.3	Internal User	Provides a valid email address (Returns to Step 4).
AF-02	Invalid Verification Code (Branches from Step 7)	
Sub step	Actor	Action
AF-02.1	System	Detects an incorrect verification code.
AF-02.2	System	Notifies the user of the error and prompts to re-provide the code.
AF-02.3	User	Provides the verification code again (Returns to Step 7).
AF-03	Verification Code Expired (Branches from Step 7)	
Sub step	Actor	Action
AF-03.1	System	Detects that the verification code has expired (Violates BR-03).
AF-03.2	System	Notifies the user of the expiration.
AF-03.3	User	Requests a new verification code (Returns to Step 4).
AF-04	Password Validation Failed (Branches from Step 10)	
Sub step	Actor	Action
AF-04.1	System	Detects a password policy violation or password mismatch (Violates BR-05, BR-06, BR-07).
AF-04.2	System	Notifies the user of the validation failure.
AF-04.3	User	Provides valid password information (Returns to Step 10).
EX-01	Delivery Service Failure (Branches from Step 4)	
Sub step	Actor	Action
EX-01.1	System	Fails to dispatch the verification email.
EX-01.2	System	Notifies the user of a temporary service error.

EX-01.3	User	Acknowledges the error and terminates the process (Use Case ends).
EX-02	Update Failure (Branches from Step 11)	
Sub step	Actor	Action
EX-02.1	System	Encounters an internal error while updating the credentials in the repository.
EX-02.2	System	Notifies the user of the error, discards changes, and aborts the process (Use Case ends).

Business Rules

#	Rule Description
BR-01	Role Authorization: Only Business Admin, Branch Manager, Inventory Manager, and Staff may access the Forgot Password feature.
BR-02	Founder Restriction: Founder accounts cannot use the Forgot Password feature and must undergo manual administrative recovery.
BR-03	Code Expiration: The verification code expires exactly 10 minutes after generation.
BR-04	Code Singularity: A verification code can only be used once.
BR-05	Password Length: The new password must contain at least 8 characters.
BR-06	Password Complexity: The new password must contain uppercase, lowercase, numeric, and special characters.
BR-07	Password History: The new password must be structurally different from the current password.
BR-08	Audit Trail: Password reset activities must be permanently recorded in the Audit Log.

3.3. Founder

3.3.1 <<UC02- Create Business Admin Account >>

3.3.1.1 Screen Mock-up

Stratos Enterprise

System Administration

Dashboard

Admin Management

Revenue Reports

Cash Flow

Profile

Settings

Log Out

System Admin Portal

Search accounts...

🔔

?

Admin Management > Create Account

Create Business Admin Account

Cancel

Create Account

👤

Personal Information

Upload Photo

JPG or PNG. Max 2MB.

Full Name

e.g. Jonathan Vickers

Email Address

j.vickers@stratos.com

Contact Number

+1 (555) 000-0000

Temporary Employee ID

STR-TEMP-000

🛡️

Security Settings

Enable 2FA by Default

Require multi-factor authentication upon first login.

Initial Password

👁️

☐ Send invite email for password setup

👤

Account Assignment

Assigned Branch

Select Branch...

Role Selector

👤

Business Admin

Full branch-level access

✓

👤

Regional Manager

Oversees multiple branches

👤

Support Specialist

Limited administrative rights

Account Summary

StatusPENDING SETUP

BranchUnassigned

RoleBusiness Admin

Once created, an activation link will be valid for 24 hours. The admin must complete registration within this period.

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Security Policy

Audit Logs

Figure 3-1: Screen Design of Create Business Admin Account

Table 3-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Upload Photo	Upload	No	2MB	Upload user avatar. Constraint: JPG or PNG format only.
2	Full Name	Text	Yes	-	Personal name of the admin (e.g., Jonathan Vickers).
3	Email Address	Text	Yes	-	Must be a valid and strictly unique email address structure.
4	Contact Number	Text	No	-	Phone contact information.

5	Temporary Employee ID	Text	No	-	Internal tracking ID (e.g., STR-TEMP-000).
6	Enable 2FA by Default	Switch	Yes	-	Requires multi-factor authentication upon first login.
7	Initial Password	Text	No	-	Manual entry of temporary password (mutually exclusive with email setup).
8	Send invite email for password setup	Checkbox	No	-	Triggers email dispatch for users to set their own password.
9	Assigned Branch	Drop-down	Yes	-	Selection of the physical/logical branch the admin is assigned to.
10	Role Selector	Radio/List	Yes	-	Authorization level selection (Business Admin, Regional Manager, Support Specialist).
11	Cancel	Button	-	-	Terminates the process without creating an account.
12	Create Account	Button	-	-	Submits the form to validate and generate the account.

3.3.1.2 Use Case Description

Use Case ID	UC02	Use Case Name	Create Business Admin Account		
Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Founder				
Description	Founder creates a new Business Admin account to grant management access to the system.				

Precondition	Founder is successfully authenticated and holds user management privileges.	
Trigger	Founder requests to create a new Business Admin account.	
Post-Condition	A new Business Admin account is generated, stored, and set to an initial active or pending state.	
Main flows		
Step	Actor	Action
1	Founder	Requests to create a new Business Admin account.
2	System	Prompts for the necessary new account details (e.g., Full Name, Email, Contact Number).
3	Founder	Provides the requested account information.
4	System	Validates the provided information against BR-01 and BR-02.
5	System	Generates the new account, securely records the credentials, and notifies the Founder of the successful creation.

Alternative flows

4a	Information validation fails (Violates BR-01)	
Sub step	Actor	Action
4a.1	System	Notifies the Founder of the specific errors (e.g., invalid format, missing required details) and prompts to re-provide the information.
4a.2	Founder	Provides the corrected information (Returns to step 4).
4b	Email duplication (Violates BR-02)	
Sub step	Actor	Action
4b.1	System	Notifies the Founder of the duplication and prompts for a different email address.
4b.2	Founder	Provides a different email address (Returns to step 4).
*a	Cancellation (Can occur at any step before step 5)	
Sub step	Actor	Action
*a.1	Founder	Requests to cancel the account creation operation.
*a.2	System	Terminates the creation process and returns to the previous state.

Business Rules

#	Rule Description
----------	-------------------------

BR-01	Account details must contain valid formats (e.g., valid email structure, minimum length for names, no strictly forbidden special characters).
BR-02	The email address associated with the new account must be absolutely unique across the entire system.
BR-03	The system must automatically generate a secure temporary password and dispatch it via email to the newly registered Business Admin.

3.3.2 <<UC27- Update Business Admin Account Profile>>

3.3.2.1 Screen Mock-up

Stratos Enterprise

SYSTEM ADMINISTRATION

Dashboard

Admin Management

Revenue Reports

Cash Flow

Profile

Settings

Log Out

System Admin Portal

Search parameters...

Admin User

Admin Management > Update Account

Update Business Admin Account

Configure user profile, permissions, and security settings.

Cancel

Save Changes

Personal Information

Change Photo

Full Name

Robert Smith

Email Address

robert.smith@stratos.cr

Contact Number

+1 (555) 234-5678

Employee ID

SE-99823

Account Assignment

Assigned Branch

Downtown Branch

Role Selector

Business Admin

Controls regional data access filters.

Defines system-wide permissions.

Recent Account Activity

View Logs

Success Login from IP 192.168.1.42

Today at 08:42 AM • Chrome on MacOS

Updated Branch Inventory Permissions

Yesterday at 04:15 PM • Admin Panel

Security & Access

Account Status

Active users can access the system

2-Factor Authentication

Mandatory for Business Admin roles

Reset Password

User will receive a secure link to reset their credentials.

System Hierarchy Map

This user sits at the Branch Admin level with cross-regional reporting access.

Danger Zone

CRITICAL WARNING

Deactivating this account will immediately revoke all access and pause all active background processes assigned to Robert Smith.

Deactivate Account

Figure 3-2: Screen Design of Update Business Admin Account Profile

Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description

1	Profile Photo	Upload	No	-	Allows uploading a new avatar image. Subject to file size/type constraints (BR-03).
2	Change photo	Button	-	-	Triggers the file selection dialog for photo upload.
3	Remove photo	Button	-	-	Clears the current photo and resets to default avatar.
4	Business Admin's Fullname	Text	Yes	50	Standard text input for full name. Must not contain numbers/special chars (BR-01).
5	Username	Text	Yes	30	Account login name. Must be system-wide unique.
6	Email Address	Text	Yes	100	Contact email. Must follow valid email format and be unique (BR-02).
7	Company Name	Text	Yes	100	Name of the organization the admin belongs to.
8	Phone Number	Text	No	15	Contact number. Must follow valid phone formatting (BR-01).
9	Assigned Role	Drop-down	Yes	-	Selects the authorization level (e.g., Business Admin).
10	Status (Active)	Toggle	Yes	-	Switches account state between Active and Inactive.
11	Cancel	Button	-	-	Discards changes and returns to previous state (Alt flow *a).
12	Save Changes	Button	-	-	Submits data for validation and update processing (Step 4 & 5).

3.3.2.2 Use Case Description

Use Case ID	UC27	Use Case Name		Update Business Admin Account Profile	
Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Founder				
Description	Founder updates the profile information of an existing Business Admin account to ensure data accuracy.				
Precondition	PRE-01: Founder is successfully authenticated. PRE-02: The target Business Admin account must already exist in the system.				
Trigger	TRG-01: Founder requests to update a specific Business Admin account.				
Post-Condition	POS-01: The targeted Business Admin account's profile is successfully updated and saved in the system.				
Main flows					
Step	Actor		Action		
1	Founder		Requests to view the details of a specific Business Admin account for updating.		
2	System		Retrieves and displays the current profile details of the selected account.		
3	Founder		Provides the updated profile information (e.g., Full Name, Email, Contact Number).		
4	System		Validates the updated information against BR-01 and BR-02.		
5	System		Updates the account record and notifies the Founder of the successful update.		

Alternative flows

2a	Account unavailable (Account deleted or deactivated by another process)	
Sub step	Actor	Action
2a.1	System	Notifies the Founder that the requested account no longer exists or is unavailable.
2a.2	System	Terminates the update process and returns to the previous state.
4a	Information validation fails (Violates BR-01)	
Sub step	Actor	Action
4a.1	System	Notifies the Founder of the specific errors (e.g., invalid email format) and prompts to correct the data.
4a.2	Founder	Provides the corrected information (Returns to step 4).
4b	Email duplication (Violates BR-02)	
Sub step	Actor	Action
4b.1	System	Notifies the Founder that the updated email already belongs to another user.
4b.2	Founder	Provides a different, unique email address (Returns to step 4).
*a	Cancellation (Can occur at any step before step 5)	
Sub step	Actor	Action
*a.1	Founder	Requests to cancel the update operation.
*a.2	System	Discards any unsaved changes, terminates the process, and returns to the previous state.

Business Rules

#	Rule Description
BR-01	Updated profile details must adhere to system data formats (e.g., valid email structure, valid phone number format, no forbidden special characters).

BR-02	If the email address is updated, the new email must be absolutely unique across the entire system (excluding the current target account's existing email).
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3.3.3 <<UC28- View Business Admin Account Details>>

3.3.3.1 Screen Mock-up

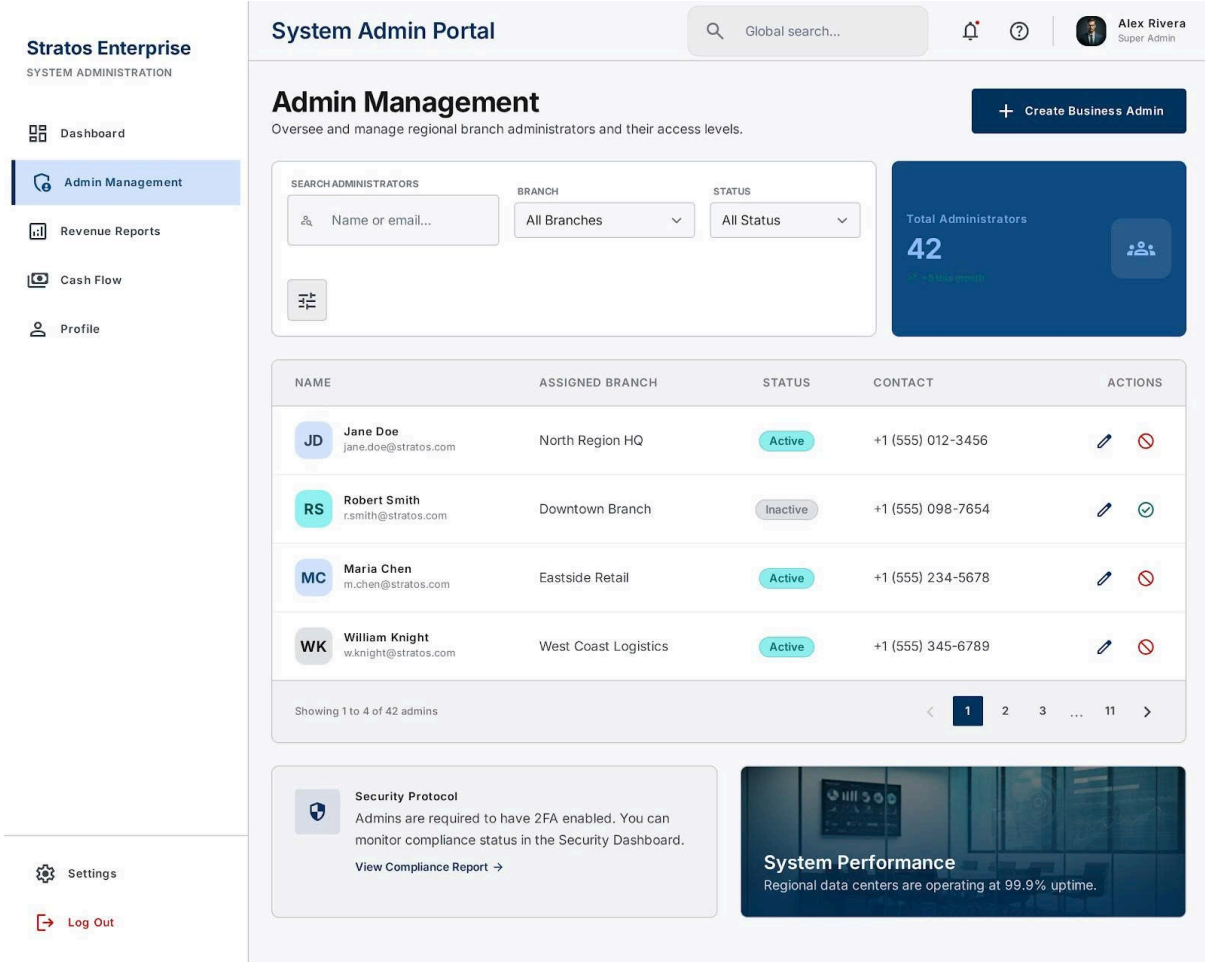


Figure 3-3: Screen Design of View Business Admin Account Details

Table 3-3: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Profile Avatar	Image	No	-	Displays the uploaded avatar of the Business Admin.
2	Admin ID	Text (Read-only)	Yes	-	Unique identifier of the account, displayed for reference.

3	Full Name	Text (Read-only)	Yes	-	Displays the registered full name.
4	Email Address	Text (Read-only)	Yes	-	Displays the contact email. Masked if dictated by BR-01.
5	Assigned Role	Text (Read-only)	Yes	-	Shows specific administrative privileges granted.
6	Account Status	Status Badge	Yes	-	Visual indicator of active/inactive state, fetched in real-time (BR-02).
7	Back	Button	-	-	Returns Actor to the previous view/state.

3.3.3.2 Use Case Description

Use Case ID	UC28	Use Case Name		View Business Admin Account Details	
Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Founder				
Description	Founder views the detailed profile and current status of a specific Business Admin account.				
Precondition	PRE-01: Founder is successfully authenticated and holds the required permissions to view account details.				
Trigger	TRG-01: Founder requests to view the detailed information of a specific Business Admin account.				
Post-Condition	POS-01: The detailed profile of the selected Business Admin account is retrieved and successfully displayed without exposing highly sensitive credentials.				
Main flows					

Step	Actor	Action
1	Founder	Requests to view the details of a specific Business Admin account.
2	System	Validates the existence of the account and retrieves the most up-to-date account details, masking sensitive information according to BR-01 and BR-02.
3	System	Displays the comprehensive profile and current active/inactive status of the requested account.

Alternative flows

2a	Account unavailable (Account was permanently deleted or access is restricted)	
Sub step	Actor	Action
2a.1	System	Notifies the Founder that the requested account no longer exists or is unavailable for viewing.
2a.2	System	Terminates the view process and returns to the previous state.

Business Rules

#	Rule Description
BR-01	Data Masking: The system must strictly exclude or mask highly sensitive data (e.g., hashed passwords, security tokens) when transmitting and displaying account details.
BR-02	Real-time Fetching: The retrieved data must reflect the real-time status of the account directly from the core database to prevent displaying stale or cached information.

3.3.4 <<UC29- Deactivate Business Admin Account>>

3.3.4.1 Screen Mock-up

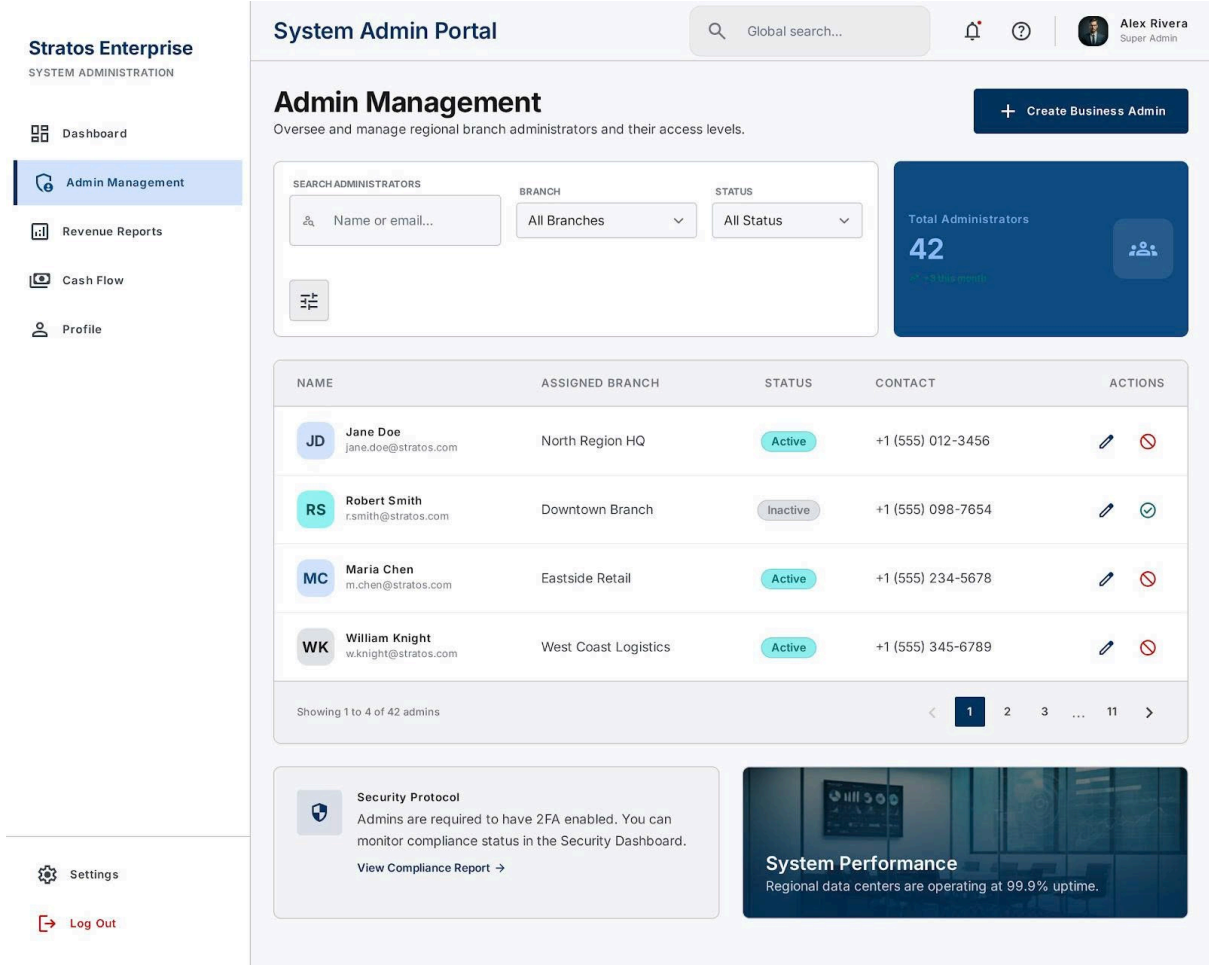


Figure 3-4: Screen Design of Deactivate Business Admin Account

Table 3-4: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Account Information	Text (Read-only)	-	-	Displays target Business Admin details (ID, Name, Email) for verification (mapped to Main flow Step 2).
2	Deactivation Reason	Textarea	Yes	255	Captures the reason for account deactivation. Must be validated by BR-03.

3	Confirm Deactivation	Button	-	-	Submits the confirmation (Main flow Step 3) and triggers execution.
4	Cancel	Button	-	-	Triggers Alternative flow *a (Cancellation) and closes the prompt.

3.3.4.2 Use Case Description

Use Case ID	UC29	Use Case Name		Deactivate Business Admin Account	
Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Founder				
Description	Founder securely disables an existing Business Admin account, preventing any further access or operations by that user without deleting their historical data.				
Precondition	PRE-01: Founder is successfully authenticated. The target Business Admin account must exist and be currently in an "Active" state.				
Trigger	TRG-01: Founder requests to deactivate a specific Business Admin account.				
Post-Condition	POS-01: The target account's status is set to "Deactivated", active sessions are terminated, and the action is permanently logged.				
Main flows					
Step	Actor		Action		
1	Founder		Requests to deactivate a specific Business Admin account.		
2	System		Retrieves the account status, verifies it is active, and prompts the Founder for a final confirmation along with a reason for deactivation.		

3	Founder	Provides the confirmation and the deactivation reason.
4	System	Executes the deactivation process securely according to BR-01 and BR-02.
5	System	Updates the account status to "Deactivated" and notifies the Founder of the successful operation.

Alternative flows

2a	Account Unavailable or Already Deactivated (Concurrency scenario)	
Sub step	Actor	Action
2a.1	System	Notifies the Founder that the target account is either not found or already in a "Deactivated" state.
2a.2	System	Terminates the process and returns to the previous state.
*a	Cancellation (Can occur at any step prior to execution)	
Sub step	Actor	Action
*a.1	Founder	Requests to cancel the deactivation process (e.g., declines the confirmation).
*a.2	System	Terminates the process without making any changes and returns to the previous state.

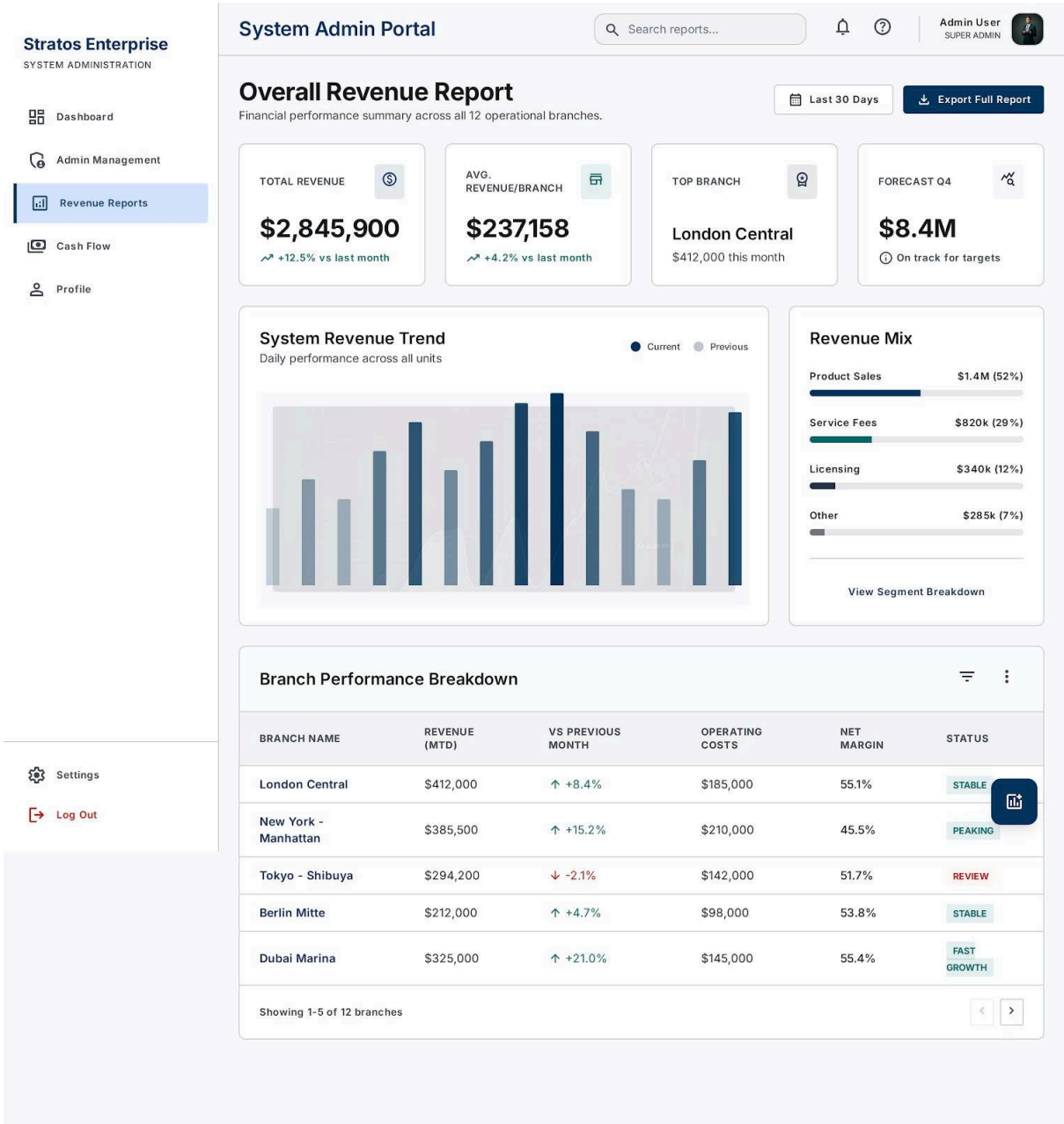
Business Rules

#	Rule Description
BR-01	Audit Trail Compliance: The system must record the deactivation event, including the exact timestamp, the acting Founder's ID, the target Business Admin's ID, and the provided reason, into a secure, immutable audit log.

BR-02	Immediate Session Revocation: Upon changing the account status to "Deactivated", the system must instantaneously invalidate all active authentication tokens (e.g., JWT) and forcefully terminate any current sessions of the target Business Admin.
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3.3.5 <<UC23- Generate Revenue Reports>>

3.3.5.1 Screen Mock-up



#	Field Name	Type	Mandatory	Max Length	Description
1	From Date	Date Picker	Yes	-	Starting date for the report. Must be $\leq$ To Date and $\leq$ Current Date (BR-01).
2	To Date	Date Picker	Yes	-	Ending date for the report. Must follow chronological validation (BR-01).
3	Branch Selection	Drop-down	Yes	-	Select a specific branch or "Entire Chain". Determines data aggregation scope (BR-02).
4	Generate Report	Button	-	-	Triggers the system to validate inputs and retrieve aggregated data.
5	Export to CSV/PDF	Button	-	-	Downloads the generated report to local storage (Noted missing in Use Case).
6	Revenue Metrics Panel	Text / Label	-	-	Read-only display of total revenue and order count based on "Completed" or "Paid" status (BR-02).
7	Revenue Chart	Data Visualization	-	-	Graphical representation (e.g., Bar/Line chart) of the aggregated financial data.

3.3.5.2 Use Case Description

Use Case ID	UC23	Use Case Name	Generate Revenue Report
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Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Founder				
Description	Founder generates and views total revenue data and charts for the entire pharmacy chain, utilizing specific filtering criteria such as date ranges and branch selections.				
Precondition	PRE-01: Founder is successfully authenticated and holds financial reporting privileges.				
Trigger	TRG-01: Founder requests to generate a revenue report.				
Post-Condition	POS-01: The system successfully aggregates the financial data and displays the revenue metrics and charts based on the requested criteria.				
Main flows					
Step	Actor		Action		
1	Founder		Requests to generate a revenue report.		
2	System		Prompts the Founder for report filtering criteria (e.g., Date Range, Specific Branches or Entire Chain).		
3	Founder		Provides the desired filtering criteria.		
4	System		Validates the provided filtering criteria against BR-01.		
5	System		Retrieves, aggregates the financial data according to BR-02, and generates the corresponding charts and metrics.		
6	System		Displays the comprehensive revenue report to the Founder.		

Alternative flows

4a	Invalid Filtering Criteria (Violates BR-01)				
Sub step	Actor	Action			

4a.1	System	Notifies the Founder of the invalid criteria (e.g., 'From Date' is greater than 'To Date') and prompts to re-provide the parameters.
4a.2	Founder	Provides the corrected filtering criteria (Returns to step 4).
5a	No Data Found	
Sub step	Actor	Action
5a.1	System	Identifies that there are no recorded transactions matching the provided criteria.
5a.2	System	Displays an empty report with zero totals and notifies the Founder that no data exists for the selected parameters.

Business Rules

#	Rule Description
BR-01	Date Range Validity: The 'From Date' must be chronologically less than or equal to the 'To Date', and the 'To Date' cannot exceed the current system date.
BR-02	Revenue Aggregation Logic: Total revenue must be calculated exclusively from orders/invoices that possess a "Completed" or "Paid" status within the selected timeframe and branch. Canceled, refunded, or pending orders must be strictly excluded from the revenue calculation.

3.3.6 <<UC24- Generate Cash Flow Statistics>>

3.3.6.1 Screen Mock-up

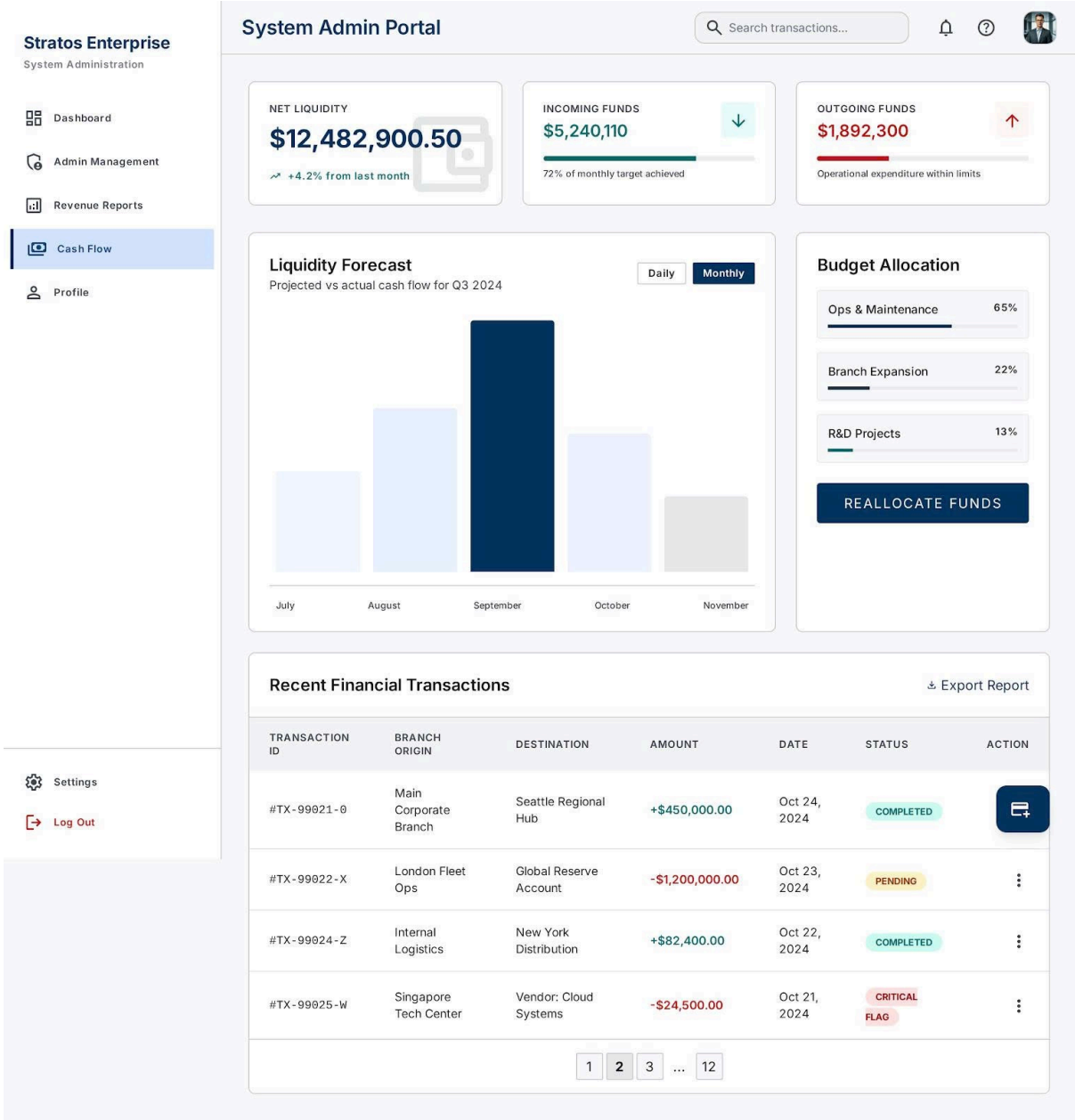


Figure 3-6: Screen Design of Generate Cash Flow Statistics

Table 3-6: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	From Date	Date Picker	Yes	-	Start date for the cash flow statistics. Must comply with BR-01 (<= To Date, no future dates).

2	To Date	Date Picker	Yes	-	End date for the cash flow statistics. Must comply with BR-01.
3	Branch	Drop-down	No	-	Allows filtering by a specific branch. Defaults to 'All Branches' if left blank.
4	Generate	Button	-	-	Triggers the validation and calculation logic (BR-01, BR-02).
5	Total Inflows	Label / Text	-	-	Read-only display of total successful sales revenues (per BR-02).
6	Total Outflows	Label / Text	-	-	Read-only display of total completed payments and costs (per BR-02).
7	Net Cash Flow	Label / Text	-	-	Read-only display of Profit/Loss (Inflows - Outflows) (per BR-02).
8	Export Data	Button	-	-	Optional action to download the generated statistics as a file (CSV/PDF).

3.3.6.2 Use Case Description

Use Case ID	UC24	Use Case Name		Generate Cash Flow Statistics	
Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Founder				
Description	Founder generates and views the system's cash flow statistics, tracking actual profit or loss based on recorded financial inflows and outflows.				

Precondition	PRE-01: Founder is successfully authenticated and holds financial reporting privileges.	
Trigger	TRG-01: Founder requests to view the cash flow statistics.	
Post-Condition	POS-01: The system aggregates financial data, performs cash flow calculations, and successfully displays the profit/loss statistics.	
Main flows		
Step	Actor	Action
1	Founder	Requests to generate the cash flow statistics.
2	System	Prompts for the cash flow filtering criteria (e.g., Time Period, Branch).
3	Founder	Provides the desired filtering criteria.
4	System	Validates the provided criteria against BR-01.
5	System	Retrieves financial data and calculates the net cash flow (profit or loss) according to BR-02.
6	System	Displays the detailed cash flow statistics, including total inflows, outflows, and net profit/loss.

Alternative flows

4a	Invalid Filtering Criteria (Violates BR-01)	
Sub step	Actor	Action
4a.1	System	Notifies the Founder of the specific criteria errors (e.g., invalid date range) and prompts for correction.
4a.2	Founder	Provides the corrected filtering criteria (Returns to step 4).
5a	No Financial Data Found	
Sub step	Actor	Action

5a.1	System	Identifies that there are no recorded inflows or outflows for the specified period and branch.
5a.2	System	Displays an empty cash flow statement with zero values and notifies the Founder of the lack of data.

Business Rules

#	Rule Description
BR-01	Time Period Validity: The start date for the statistics must be prior to or equal to the end date, and future dates are strictly prohibited.
BR-02	Cash Flow Calculation Formula: - Total Inflows = Sum of all successful sales revenues. - Total Outflows = Sum of all completed supplier payments + documented operating costs. - Net Cash Flow (Profit/Loss) = Total Inflows - Total Outflows. Only finalized and successfully paid transactions are included in this calculation.

3.3.7 <<UC25- Export Financial Reports>>

3.3.7.1 Screen Mock-up

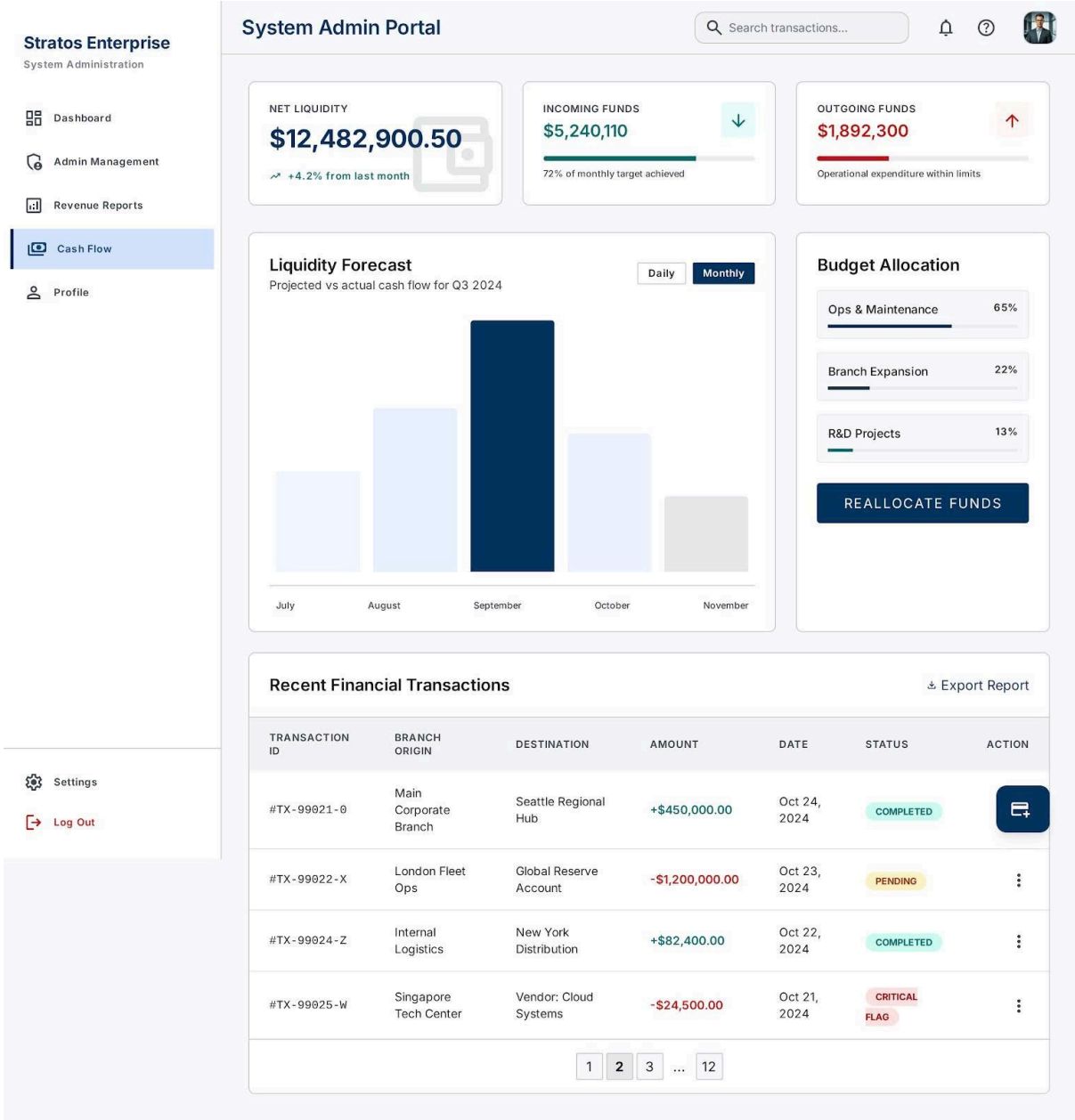


Figure 3-7: Screen Design of Export Financial Reports

Table 3-7: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Report Type	Drop-down	Yes	-	Select the classification of the financial report (Revenue, Cash Flow, Expenses).

2	Start Date	Date Picker	Yes	-	The beginning date for the report's data scope.
3	End Date	Date Picker	Yes	-	The ending date for the report's data scope.
4	Branch	Drop-down	Yes	-	Select the specific branch to filter the financial data.
5	Include Pending Transactions?	Checkbox	No	-	Flag to include uncommitted/pending financial records in the export.
6	Export Format	Radio Button	Yes	-	Selection of the output file format (PDF, Excel, CSV).
7	Password Protect File?	Toggle / Checkbox	No	-	Flag to enable encryption/password protection on the generated file.
8	Cancel	Button	-	-	Aborts the export request and dismisses the prompt.
9	Generate & Download	Button	-	-	Triggers the system to validate inputs, generate the file, and initiate the download.

3.3.7.2 Use Case Description

Use Case ID	UC25	Use Case Name		Export Financial Reports	
Author	Trần Trung Hiếu	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Founder				
Description	Founder requests to generate and export specific financial reports (e.g., Revenue, Cash Flow) into standard formats for offline archiving and sharing.				

Precondition	PRE-01: Founder is successfully authenticated and holds specific privileges to export financial data. The requested financial data must exist in the system.	
Trigger	TRG-01: Founder requests to export a financial report.	
Post-Condition	POS-01: The system securely provides the requested report file to the Founder and logs the export event.	
Main flows		
Step	Actor	Action
1	Founder	Requests to export a financial report.
2	System	Prompts the Founder for export criteria (Report Type, Date Range, Branch, and Export Format).
3	Founder	Provides the required export criteria.
4	System	Validates the criteria against BR-01 and confirms data availability for the selected scope.
5	System	Retrieves the data, formats the report according to BR-02, and generates the file.
6	System	Provides the generated file to the Founder and records the export action according to BR-03.

Alternative flows

4a	No Data Found for Export	
Sub step	Actor	Action
4a.1	System	Identifies that there are no financial records matching the provided criteria.
4a.2	System	Notifies the Founder that the export cannot proceed due to empty data and prompts to adjust the criteria.

4a.3	Founder	Provides the adjusted export criteria (Returns to step 4).
5a	File Generation Failure	
Sub step	Actor	Action
5a.1	System	Encounters an internal error during the report formatting or file generation process.
5a.2	System	Notifies the Founder of the generation failure and safely aborts the operation without logging a successful export.

Business Rules

#	Rule Description
BR-01	Data Scope Consistency: The exported data must strictly match the financial records (Completed/Paid status only) within the requested date range, explicitly excluding any uncommitted or pending transactions to ensure offline data integrity.
BR-02	Format and Meta-data Standards: The exported file must be strictly formatted as either a standard PDF document (.pdf) or an Excel spreadsheet (.xlsx). Every generated report MUST automatically embed a meta-data header containing: The exact timestamp of generation, the ID of the Founder generating the report, and the scope of the data.
BR-03	Financial Audit Trail: Every successful export of financial data must be permanently recorded in the system's audit log. The log must capture the Admin ID, Report Type, selected Date Range, Export Format, and timestamp to monitor data exfiltration.

3.4. Business Admin

3.4.1 <<UC034- View Business Analysis Report >>

3.4.1.1 Screen Mock-up

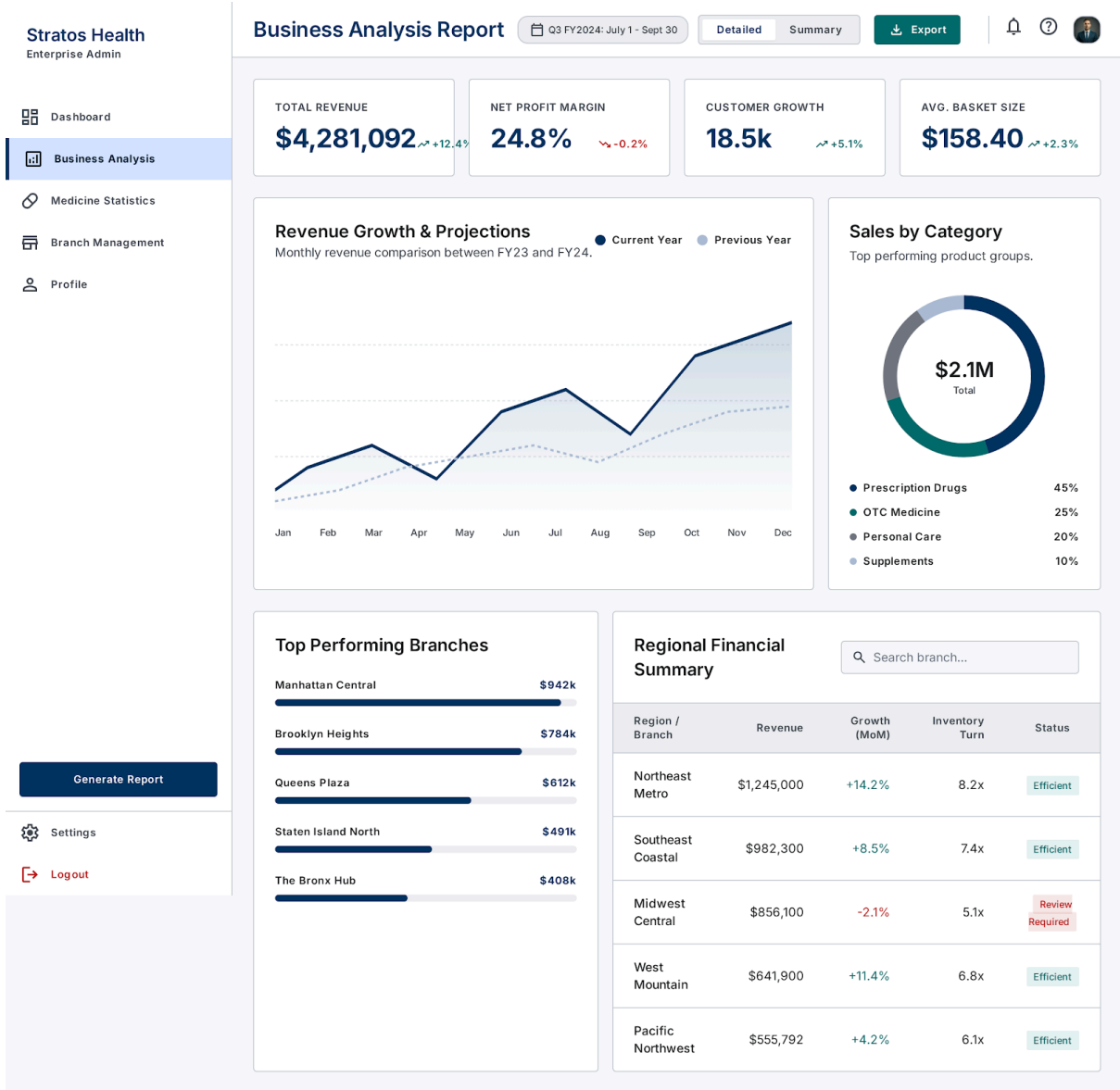


Table 3-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Date Range Filter	Date Range Picker	No	2MB	Select reporting period for business analysis.

2	Detail View Button	Button	No	-	Display detailed business analysis information.
3	Summary View Button	Button	No	-	Display summarized business analysis information.
4	Export Report	Button	No	-	Export business analysis report to Excel or PDF format.
5	Total Revenue	KPI Card	No	-	Display total revenue generated during the selected period.
6	Net Profit Margin	KPI Card	No	-	Display overall profit margin percentage.
7	Customer Growth	KPI Card	No	-	Display customer growth rate compared to the previous period.
8	Average Basket Size	KPI Card	No	-	Display average revenue per transaction.
9	Revenue Growth & Projections	Line Chart	No	-	Visualize revenue trends and projected growth.
10	Sales by Category	Doughnut Chart	No	-	Display revenue contribution by medicine category.
11	Top Performing Branches	Ranking Panel	No	-	Display branches with highest revenue performance.
12	Regional Financial Summary	Data Table	No	-	Display branch financial statistics including revenue, growth rate, inventory turnover, and status.
13	Search Branch	Search Box	No	100	Search specific branch information within the report.

14	Notification Button	Icon Button	No	-	View system notifications.
15	User profile Preview	Dropdown	No	-	Access profile settings and logout options.

3.3.3.2 Use Case Description

Use Case ID	UC34	Use Case Name		View Business Admin Account Details	
Author	Bùi Đức Lương	Version	1.0	Date	02/06/2026
Actor	Business Admin				
Description	Business Admin views and analyzes business performance statistics, revenue trends, branch performance, and medicine sales data across the pharmacy chain.				
Precondition	PRE-01: Business Admin is successfully authenticated and has permission to access business reports.				
Trigger	TRG-01: Business Admin requests to access the Business Analysis Report screen.				
Post-Condition	POS-01: Business analysis data is successfully retrieved and displayed. Reports can be exported if requested.				
Main flows					
Step	Actor		Action		
1	Business Admin		Requests to access the Business Analysis Report screen.		
2	System		Retrieves business performance data based on the selected reporting period.		
3	System		Calculates revenue statistics, customer growth, profit margin, and branch performance metrics.		

4	System	Displays KPI indicators, charts, rankings, and financial summaries.
5	Business Admin	Reviews business analysis information.
6	Business Admin	Selects filters such as date range or report type.
7	System	Refreshes and displays the updated report results.
8	Business Admin	Requests to export the report.
	System	Generates and exports the report in Excel or PDF format.

Alternative flows

2a	2a No data available for selected period	
Sub step	Actor	Action
2a.1	System	No report data is found for the
2a.2	System	Displays a notification message indicating that no data is available.
2a.3	Business Admin	Selects another reporting period.

Business Rules

#	Rule Description
BR-01	Only Business Admin users are authorized to access Business Analysis Reports.

BR-02	Report data must be filtered according to the selected reporting period.
BR-03	Revenue calculations are based only on completed transactions.
BR-04	Reports can be exported to Excel (.xlsx) or PDF format.
BR-	Financial information must only be visible to authorized users.

3.4.2 <<UC06- View Medicine Statistics >>

3.4.2.1 Screen Mock-up

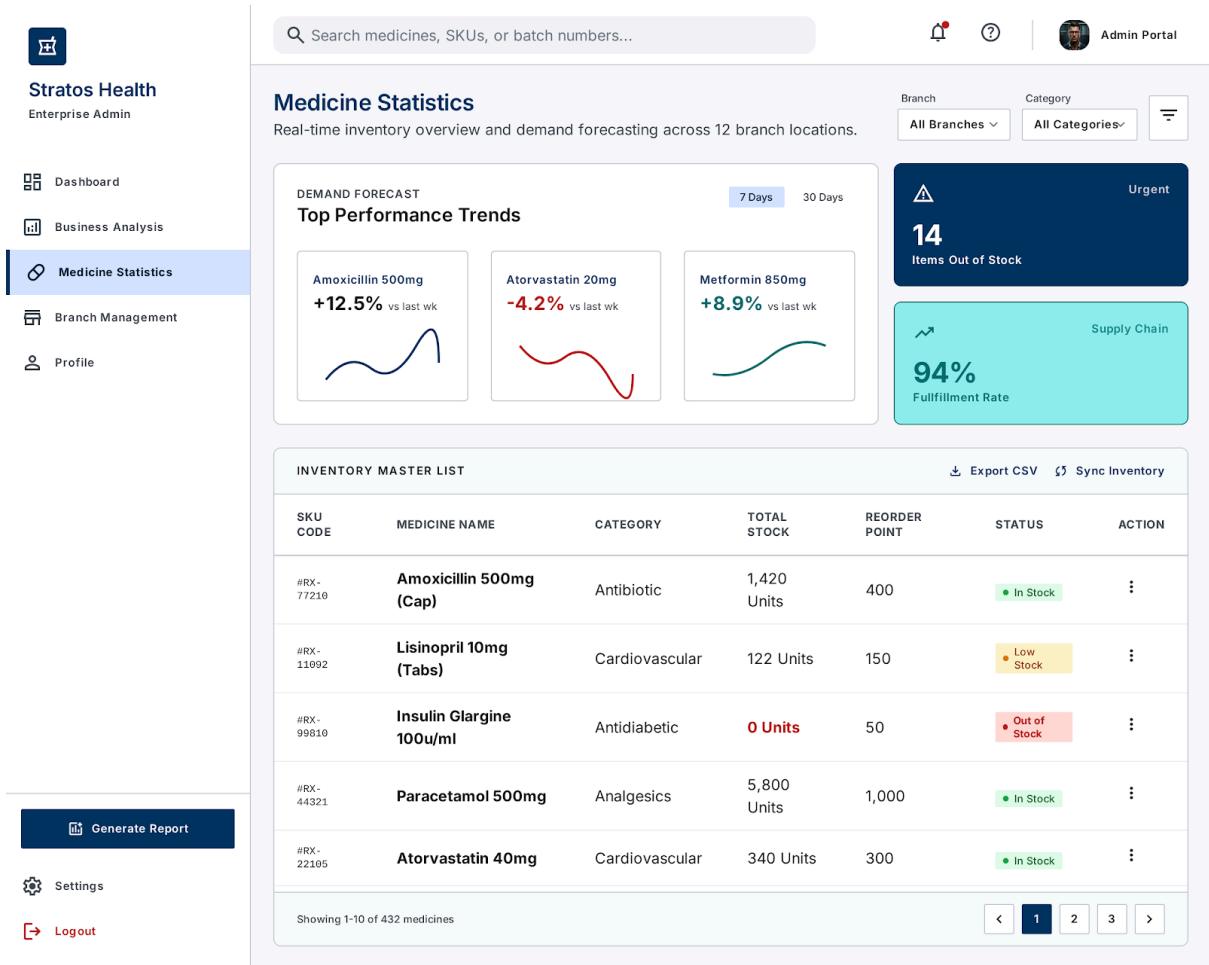


Table 3-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Search Medicine	Search Box	No		Search medicines by name, SKU code, or batch number
2	Branch Filter	Drop-down	No	-	Filter medicine statistics by branch.
3	Category Filter	Drop-down	No	-	Filter medicine statistics by medicine category.

4	Top Performance Trends	Statistics Card	No	-	Display medicine demand and sales trends.
5	Out of Stock Alert	KPI Card	No	-	Display the number of medicines currently out of stock. .
6	Fulfillment Rate	KPI Card	No	-	Display inventory fulfillment performance.
7	Inventory Master List	Data Table	No	-	Display medicine inventory information including stock quantity and reorder level.
8	Export CSV	Button	No	-	Export medicine statistics to CSV format.
9	Sync Inventory	Button	No	-	Refresh and synchronize inventory information.
10	Medicine Status	Status Label	No	-	Display revenue contribution by medicine category.
11	Action Menu	Action Button	No	-	Access detailed medicine information.

3.3.3.2 Use Case Description

Use Case ID	UC06	Use Case Name		View Medicine Statistics	
Author	Bùi Đức Lương	Version	1.0	Date	02/06/2026
Actor	Business Admin				
Description	Business Admin views medicine inventory statistics, stock levels, demand trends, and medicine performance across pharmacy branches. .				

Precondition	PRE-01: Business Admin is successfully authenticated and authorized to access medicine statistics.	
Trigger	TRG-01: Business Admin requests to access the Medicine Statistics screen.	
Post-Condition	POS-01: Medicine statistics and inventory information are successfully retrieved and displayed.	
Main flows		
Step	Actor	Action
1	Business Admin	Requests to access the Medicine Statistics screen.
2	System	Retrieves medicine inventory and sales statistics.
3	System	Displays stock levels, demand trends, inventory alerts, and medicine performance information.
4	Business System	Reviews medicine statistics and optionally applies filters.
5	Admin	Refreshes and displays updated medicine statistics
6	Business Admin	Optionally exports the statistics report.

Alternative flows

2a	2a No data available for selected period	
Sub step	Actor	Action
2a.1	System	No medicine statistics are available.
2a.2	System	Displays a notification message indicating that no data is available.

Business Rules

#	Rule Description
BR-01	Only authorized Business Admin users can access medicine statistics.
BR-02	Inventory quantities must be updated in real time across all branches.
BR-03	Medicines below the reorder point must be marked as Low Stock.
BR-04	Medicines with zero available quantity must be marked as Out of Stock.
BR-05	Statistics can be filtered by branch and medicine category.
BR-06	Reports can be exported to CSV format.

3.4.3 <<UC03- View Branch Management >>

3.4.3.1 Screen Mock-up

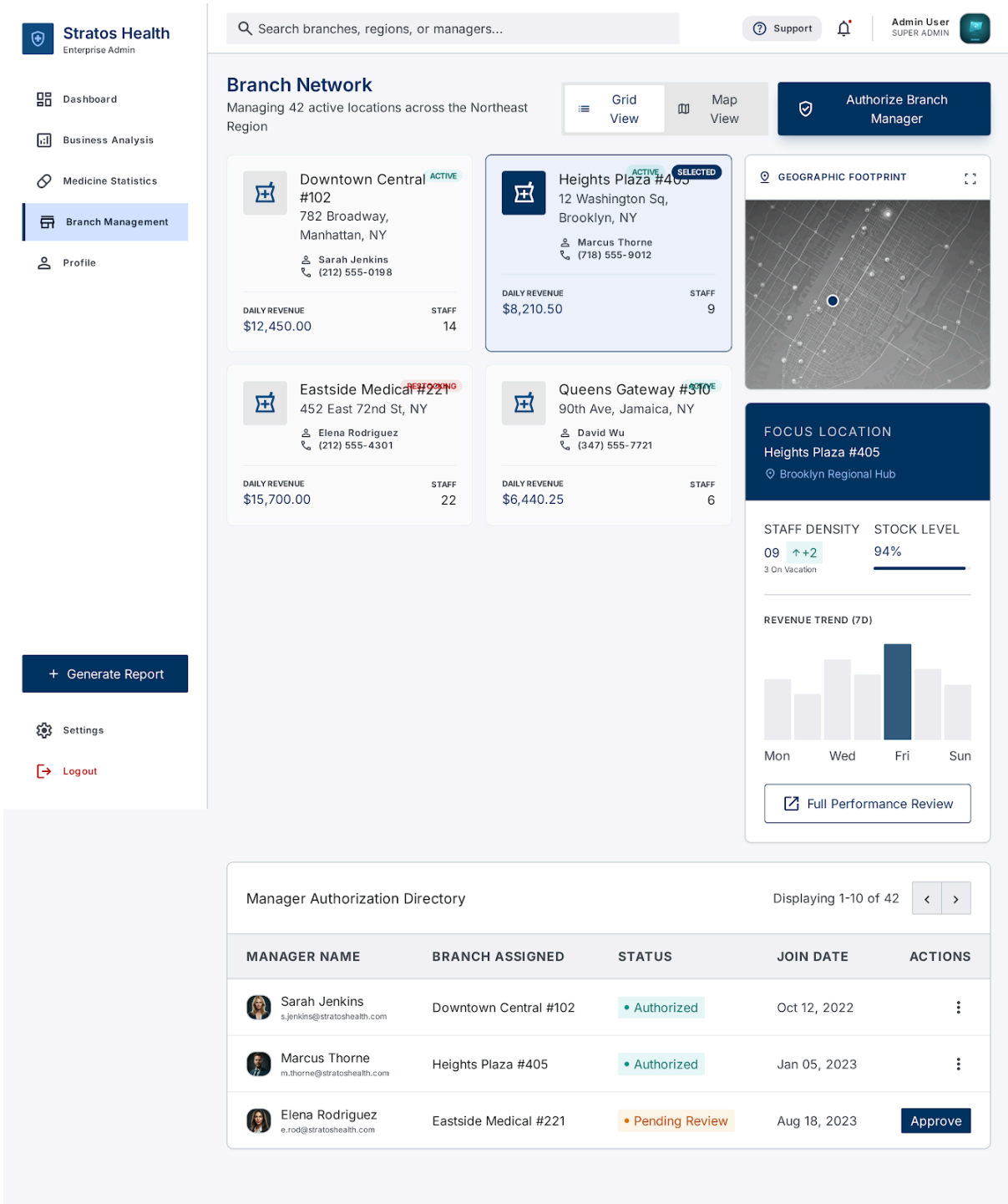


Table 3-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description

1	Report Type	Drop-down	Yes	-	Select the classification of the financial report (Revenue, Cash Flow, Expenses).
2	Start Date	Date Picker	Yes	-	The beginning date for the report's data scope.
3	End Date	Date Picker	Yes	-	The ending date for the report's data scope.
4	Branch	Drop-down	Yes	-	Select the specific branch to filter the financial data.
5	Include Pending Transactions?	Checkbox	No	-	Flag to include uncommitted/pending financial records in the export.
6	Export Format	Radio Button	Yes	-	Selection of the output file format (PDF, Excel, CSV).
7	Password Protect File?	Toggle / Checkbox	No	-	Flag to enable encryption/password protection on the generated file.
8	Cancel	Button	-	-	Aborts the export request and dismisses the prompt.
9	Generate & Download	Button	-	-	Triggers the system to validate inputs, generate the file, and initiate the download.

3.4.3.2 Use Case Description

Use Case ID	UC03	Use Case Name	Manage Pharmacy Branches
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Author	Bùi Đức Lương	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Business Admin				
Description	Business Admin manages pharmacy branch information, monitors branch performance, and authorizes branch managers.				
Precondition	PRE-01: Business Admin is successfully authenticated and holds branch management privileges (Complies with BR-01 and BR-06).				
Trigger	TRG-01: Business Admin requests to manage pharmacy branch information.				
Post-Condition	POS-01: Branch information and manager authorization data are successfully updated and securely recorded in the system.				
Main flows					
Step	Actor		Action		
1	Business Admin		Requests to manage pharmacy branch information.		
2	System		Retrieves and displays the current list of branches, performance metrics, and pending authorization requests.		
3	Business Admin		Selects a specific branch and provides updated branch information or a manager authorization decision.		
4	System		Validates the provided information and assignments against BR-02, BR-03, and BR-05.		
5	System		Saves the changes, updates the branch records, and notifies the Business Admin of the successful operation.		

Alternative flows

2a	No Branch Data Available				
Sub step	Actor	Action			

2a.1	System	Identifies that no branch records or assignments exist in the database.
2a.2	System	Displays an empty list and notifies the Business Admin that no data is available.
4a	Invalid Manager Assignment or Data	
Sub step	Actor	Action
4a.1	System	Detects invalid data input (e.g., violates BR-05 geographical linking or missing fields).
4a.2	System	Displays a validation message detailing the errors and prompts for correction.
4a.3	Business Admin	Provides the corrected information (Returns to step 4).
4b	Authorization Request Rejected	
Sub step	Actor	Action
4b.1	System	Identifies that the selected authorization request does not meet assignment requirements (e.g., violates BR-02 multiple active managers).
4b.2	System	Displays a rejection message and prevents the authorization.

Business Rules

#	Rule Description
BR-01	Access Control: Only Business Admin users can manage branch information.
BR-02	Manager Exclusivity: Each branch can only have one active Branch Manager at a time.
BR-03	Approval Workflow: Branch Manager assignments must be explicitly approved by an authorized user before activation.
BR-04	Real-time Metrics: Branch performance statistics must be fetched and updated in real time.
BR-05	Location Validity: Branch locations must be strictly linked to a valid geographic address.

BR-06	Authorization Constraint: Only explicitly authorized users can approve manager assignments.
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3.5. Branch Manager

3.5.1 <<UC09- View Branch DashBoard >>

3.5.1.1 Screen Mock-up

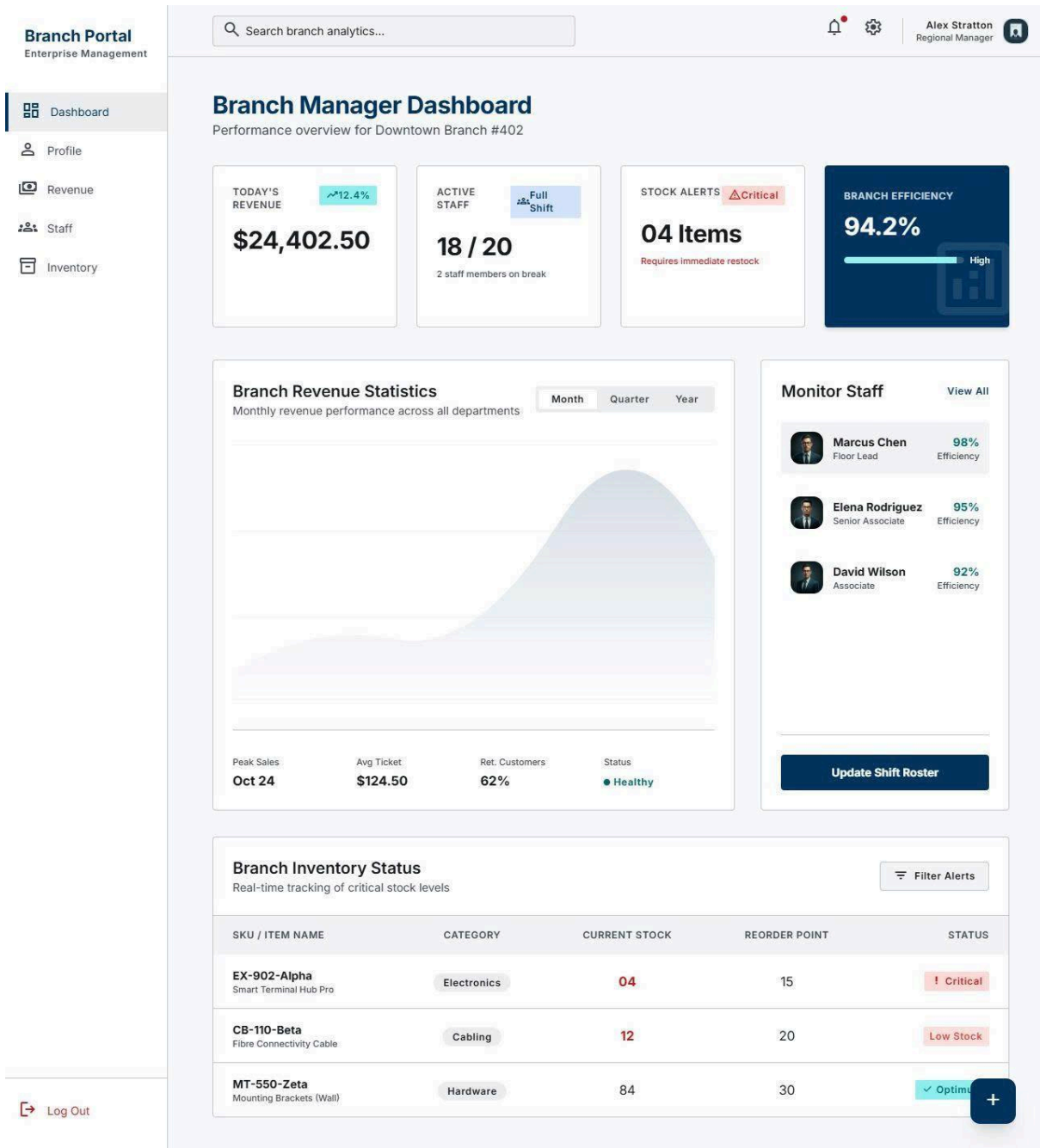


Table 3-2: Dashboard Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Search Analytics	Search Box	No	100	Search branch analytics information.
2	Today's Revenue	Card	Yes	-	Display current day's revenue.
3	Active Staff	Card	Yes	-	Display number of active staff members. .
4	Stock Alerts	Card	Yes	-	Display inventory items requiring attention.
5	Branch Efficiency	Progress Card	Yes	-	Display overall branch efficiency percentage.
6	Revenue Statistics Chart	Chart	Yes	-	Display revenue trends and performance statistics.
7	Time Period Filter	Tab Selector	No	-	Switch between Month, Quarter, and Year views.
8	Staff Monitor Panel	List	Yes	-	Display top-performing staff members and efficiency scores.
9	Update Shift Roster	Button	No	-	Open shift scheduling management screen.
10	Inventory Status Table	Table	Yes	-	Display critical inventory information and stock levels.
11	Filter Alerts	Button	No	-	Filter inventory alerts by status.
12	Add New Record	Floating Button	No	-	Create a new inventory or operational record.

3.5.1.2 Use Case Description

Use Case ID	UC09	Use Case Name		View Branch Dashboard	
Author	HanND	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Branch Manager				
Description	Branch Manager accesses the dashboard to monitor overall branch performance, revenue, staff activities, inventory alerts and confirm daily revenue.				
Precondition	PRE - 01: Branch Manager is authenticated and authorized to access branch information.				
Trigger	TRG - 01: Branch Manager selects the Dashboard menu.				
Post-Condition	POS - 01: The dashboard information is displayed successfully.				
Main flows					
Step	Actor		Action		
1	Branch Manager		Requests to access the Dashboard.		
2	System		Retrieves branch performance information.		
3	System		Displays revenue, active staff, inventory alerts, and efficiency metrics.		
4	Branch Manager		Review the displayed information.		
5	System		Maintains the dashboard for further interaction.		

Alternative flows

2a	System data unavailable	
Sub step	Actor	Action
2a.1	System	Displays an error message indicating data cannot be loaded.
2a.2	Branch Manager	Retries the request.
2a.3	System	Returns to Step 2

Business Rules

#	Rule Description
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BR-01	Dashboard data must reflect the latest branch information.
BR-02	Only authorized Branch Managers may access branch dashboards.

3.5.2 <<UC10- View Branch Revenue Statistics >>

3.5.2.1 Screen Mock-up

Branch Revenue Statistics

Search transactions...



Daily Weekly Monthly Custom

Oct 01 - Oct 24, 2023

Export Report



+12.4%

TOTAL REVENUE

\$142,500.00



-2.1%

AVERAGE TICKET

\$245.80



+8.5%

NEW CUSTOMERS

1,248



+5.3%

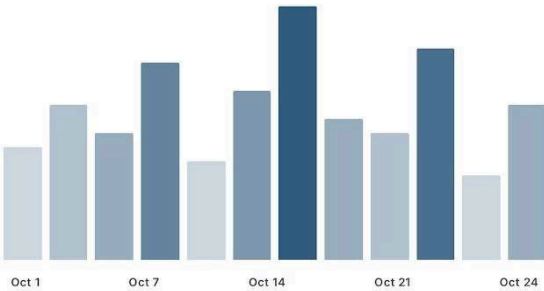
GROSS MARGIN

32.4%

Revenue Trend

Daily performance compared to previous period

Current Previous



By Category

Electronics	\$52,400 (42%)
Home Goods	\$38,100 (28%)
Apparel	\$24,300 (18%)
Groceries	\$16,200 (12%)

Top Performer
Premium Smartphones
+\$4.2k

Recent Performance by Time

View All Records

TIME BLOCK	TRANSACTIONS	REVENUE	AVG ORDER	PERFORMANCE	STATUS
08:00 - 10:00 (Opening)	42	\$8,240.00	\$196.19		Normal
11:00 - 14:00 (Peak Lunch)	128	\$32,150.00	\$251.17		High Traffic
15:00 - 17:00 (Afternoon)	64	\$14,800.00	\$231.25		Normal
18:00 - 21:00 (Evening Rush)	112	\$42,900.00	\$383.04		Critical Load

Revenue Forecast

Based on current trends, we anticipate a 15% increase in next month's revenue due to holiday seasonal factors.

ESTIMATED GROSS

\$158,000

GROWTH CONFIDENCE

92%



Optimization Insights

- Evening shift average ticket size is 25% higher. Consider increasing floor staff from 18:00 to 21:00.
- Electronics category inventory turns have slowed. Recommend promotional bundles for slow-moving SKUs.

Table 3-3: Branch Revenue Statistics Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Search Transactions	Search Box	No	100	Search revenue transactions.
2	Period Selector	Tab Selector	No	-	Select Daily, Weekly, Monthly, or Custom view.
3	Date Range Picker	Date Picker	No	-	Select reporting date range.
4	Export Report	Button	No	-	Export revenue report.
5	Total Revenue	Card	Yes	-	Display total revenue amount.
6	Average Ticket	Card	Yes	-	Display average order value.
7	Gross Margin	Card	Yes	-	Display gross profit margin percentage.
8	Revenue Trend Chart	Chart	Yes	-	Display revenue trends over time.
9	Category Revenue Panel	Summary Panel	Yes	-	Display revenue contribution by product category.
10	Performance Table	Table	Yes	-	Display detailed revenue performance records.

3.5.2.2 Use Case Description

Use Case ID	UC10	Use Case Name	View Branch Revenue Statistics
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Author	HanND	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Branch Manager				
Description	Branch Manager views branch revenue statistics, sales performance, revenue trends.				
Precondition	PRE - 01: Branch Manager is successfully authenticated and authorized to access branch revenue information.				
Trigger	TRG - 01:Branch Manager requests to view branch revenue statistics.				
Post-Condition	POS - 01: Revenue statistics and financial reports are displayed successfully.				
Main flows					
Step	Actor		Action		
1	Branch Manager		Requests to access the Branch Revenue Statistics		
2	System		Retrieves branch revenue and sales data.		
3	System		Displays revenue summaries, charts, and performance indicators.		
4	Branch Manager		Selects a reporting period or date range.		
5	System		Updates and displays the corresponding revenue statistics.		

Alternative flows

2a	Revenue data unavailable	
Sub step	Actor	Action
2a.1	System	Displays an error message indicating that revenue data cannot be retrieved.
2a.2	Branch Manager	Requests to reload the data.
2a.3	System	Returns to Step 2
4a	Invalid date range	
Sub step	Actor	Action
4a.1	System	Displays a validation message indicating the selected date range is invalid.
4a.2	Branch Manager	Selects a valid date range.

4a.3	System	Return to Step 5.
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Business Rules

#	Rule Description
BR-01	Revenue statistics must be generated from completed and approved sales transactions.
BR-02	The system shall support revenue filtering by predefined periods and custom date ranges.
BR-03	Branch Managers may only access revenue data belonging to their assigned branch.

3.5.3 <<UC32- Monitor Staff Performance >>

3.5.3.1 Screen Mock-up

Search staff members...



Alex Rivera
BRANCH MANAGER



Monitor Staff Performance

Real-time productivity, engagement, and sales metrics across your branch team.

Filter

+ New Assessment

AVERAGE SALES TARGET

84.2%

+2.4% vs last month

CUSTOMER SATISFACTION

4.8/5



TEAM ATTENDANCE

96.8%

-0.5% vs last month

TOP PERFORMER

Marcus Chen

Achieved 124% of quarterly goals with 0 unplanned absences.

[View Analysis](#)

Staff Performance Matrix

Q3 Active

12 Employees

TEAM MEMBER	SALES TARGET	TARGET PROGRESS	CUSTOMER RATING	ATTENDANCE	ACTIONS
 Sarah Jenkins Senior Advisor	\$12,500 / \$15k	83%	4.9  +0.2	98%	
 Marcus Chen Lead Specialist	\$22,400 / \$18k	124%	5.0  MAX	100%	
 David Miller Customer Associate	\$6,200 / \$10k	62%	4.2  -0.4	84%	
 Elena Rodriguez Product Expert	\$14,800 / \$15k	98%	4.7  STABLE	99%	

Showing 1-4 of 12 employees



Trend Analysis

Quarterly Growth

Net sales productivity across all roles

+12.4%
vs Q2



Manager
Insight:

Sales performance is peaking during mid-week shifts. Consider adjusting staff rotation to maximize high-conversion hours.

Recent Feedback

To: Elena Rodriguez

2h ago

"Excellent handling of the inventory discrepancy yesterday. Your attention to detail saved us a significant amount in audit time."

To: Marcus Chen

Yesterday

"Marcus continues to lead the team by example. His customer satisfaction scores are at an all-time high for this branch."

[VIEW ALL REVIEWS](#)

Table 3-4: Monitor Staff Performance Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Search Staff Members	Search Box	No	100	Search employee records.
2	Filter Staff	Button	No	-	Apply staff performance filters.
3	New Assessment	Button	No	-	Create a new employee evaluation.
4	Average Sales Target	Card	Yes	-	Display average sales target achievement.
5	Customer Satisfaction	Card	Yes	-	Show average customer satisfaction rating.
6	Team Attendance	Card	Yes	-	Display attendance percentage of team members.
7	Top Performer Panel	Information Card	Yes	-	Highlight the best-performing employee.
8	Staff Performance Matrix	Table	Yes	-	Display employee performance metrics and ratings.
9	Trend Analysis	Chart	Yes	-	Show staff productivity growth trends.
10	Manager Insight	Information Panel	No	-	Display management recommendations and insights.
11	Recent Feedback	List	No	-	Show recent employee feedback records.
12	View All Reviews	Button	No	-	Navigate to complete feedback history.

3.5.3.2 Use Case Description

Use Case ID	UC32	Use Case Name		Monitor Staff Performance	
Author	HanND	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Branch Manager				
Description	Branch Manager monitors staff activities, attendance, productivity, sales performance, and other performance indicators within the assigned branch.				
Precondition	PRE - 01: Branch Manager is successfully authenticated and authorized to access staff performance information.				
Trigger	TRG - 01:Branch Manager requests to view staff performance information.				
Post-Condition	POS - 01: Staff performance data is displayed successfully.				
Main flows					
Step	Actor		Action		
1	Branch Manager		Requests to access the Monitor Staff Performance screen.		
2	System		Retrieves staff performance records and metrics.		
3	System		Displays staff performance information, attendance records, sales achievements, and performance indicators.		
4	Branch Manager		Reviews performance information and applies filters if necessary.		
5	System		Updates and displays the corresponding staff performance data.		

Alternative flows

2a	Staff performance data unavailable	
Sub step	Actor	Action
2a.1	System	Displays an error message indicating that staff performance data cannot be retrieved.
2a.2	Branch Manager	Requests to reload the data.
2a.3	System	Returns to Step 2
4a	No matching records found	
Sub step	Actor	Action

4a.1	System	Displays a message indicating that no matching staff records were found.
4a.2	Branch Manager	Modifies the filter criteria.
4a.3	System	Return to Step 4.

Business Rules

#	Rule Description
BR-01	Branch Managers may only access staff performance data belonging to their assigned branch.
BR-02	Staff performance indicators shall be calculated based on attendance, sales achievements, and operational activities.
BR-03	The system shall provide filtering and searching capabilities for staff performance records.

3.5.4 <<UC33- Monitor Branch Inventory>>

3.5.4.1 Screen Mock-up

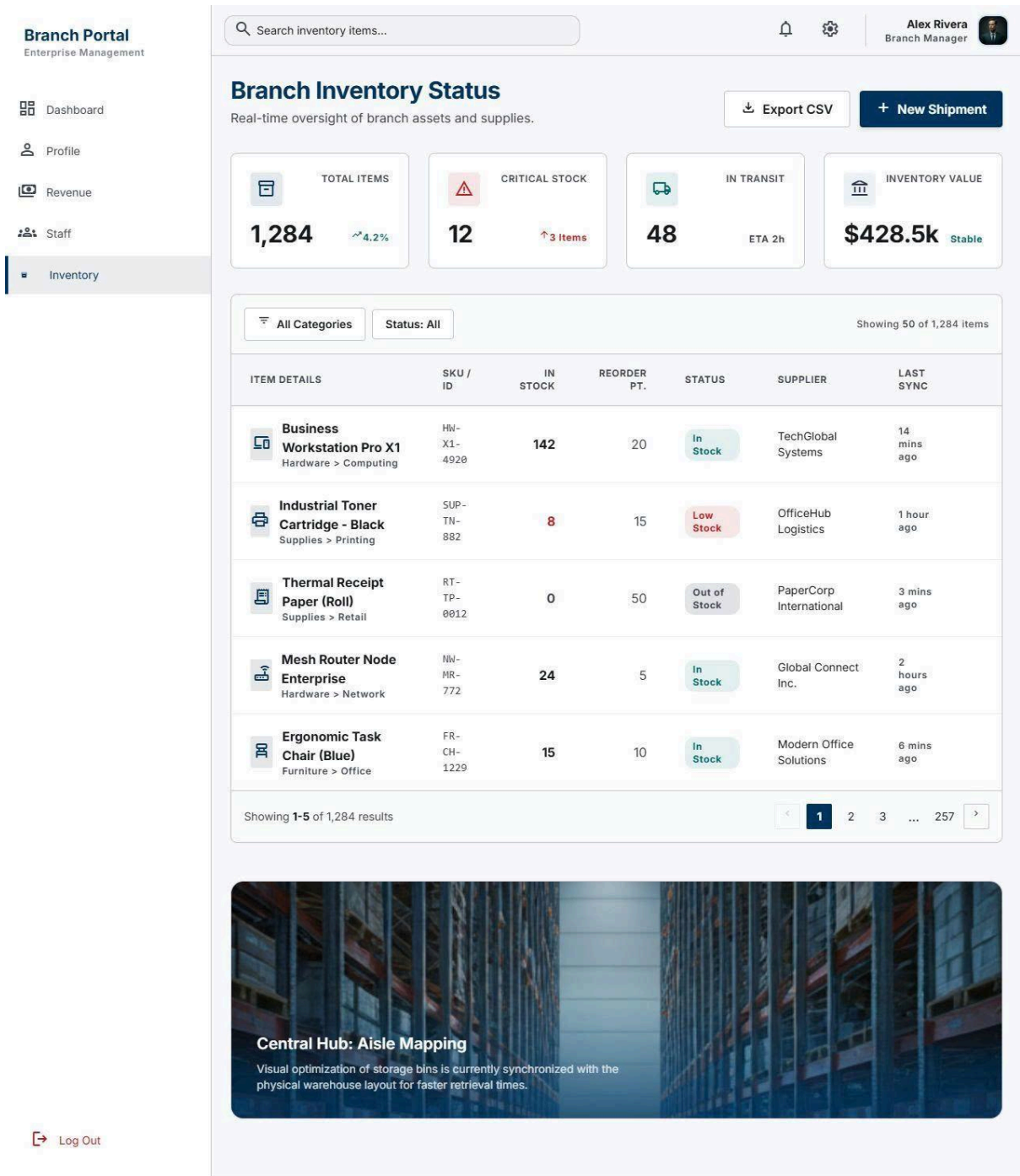


Table 3-5: Branch Inventory Status Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description

1	Search Inventory Items	Search Box	No	100	Search inventory products by name or SKU.
2	Total Items	Card	No	-	Display total inventory item count.
3	Critical Stock	Card	Yes	-	Display number of critical stock items.
4	In Transit	Card	Yes	-	Show quantity of incoming inventory shipments.
5	Inventory Value	Card	Yes	-	Display total inventory value.
6	Export CSV	Button	No	-	Export inventory data to CSV format.
7	New Shipment	Button	No	-	Create a new shipment record.
8	Category Filter	Drop-down	No	-	Filter inventory by category.
9	Status Filter	Drop-down	No	-	Filter inventory by stock status.
10	Inventory Table	Table	Yes	-	Display item details, stock levels, supplier information, and status.
11	Pagination	Navigation	No	-	Navigate between inventory pages.

3.5.4.2 Use Case Description

Use Case ID	UC33	Use Case Name	Monitor Branch Inventory		
Author	HanND	Versio n	1.0	Date	02/06/2026
Actor	Primary Actor: Branch Manager				

Description	Branch Manager monitors inventory status, stock availability, low-stock conditions, and replenishment requirements within the assigned branch.	
Precondition	PRE - 01:Branch Manager is successfully authenticated and authorized to access branch inventory information.	
Trigger	TRG - 01:Branch Manager requests to monitor branch inventory information.	
Post-Condition	POS - 01: Inventory information is displayed successfully.	
Main flows		
Step	Actor	Action
1	Branch Manager	Requests to access the Monitor Branch Inventory screen.
2	System	Retrieves inventory records of the assigned branch.
3	System	Displays inventory information, stock quantities, stock status, and low-stock alerts
4	Branch Manager	Searches, filters, or reviews inventory information.
5	System	Updates and displays the corresponding inventory data.

Alternative flows

2a	Inventory data unavailable	
Sub step	Actor	Action
2a.1	System	Displays an error message indicating that inventory data cannot be retrieved.
2a.2	Branch Manager	Requests to reload the data.
2a.3	System	Returns to Step 2
4a	No matching inventory records found	
Sub step	Actor	Action
4a.1	System	Displays a message indicating that no matching inventory records were found.
4a.2	Branch Manager	Modifies the search or filter criteria.
4a.3	System	Return to Step 4.

Business Rules

#	Rule Description
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BR-01	Branch Managers may only access inventory information belonging to their assigned branch.
BR-02	Inventory quantities shall reflect the latest approved stock transactions..
BR-03	The system shall support searching and filtering inventory records by item name, category, and stock status.

3.6. Staff

3.6.1 <<UC16-View Medicine Information>>

3.6.1.1 Screen Mock-up

Stratos Enterprise

Pharmacy Branch 042

+ New Prescription

Dashboard

Medicine Search

Invoices

Profile

System Status

Log Out

Quick search medicine, SKU or invoice...

Dr. Sarah Chen
SENIOR PHARMACIST

Medicine Search

Search the global formulary and manage local branch inventory.

Category

dosage form

Availability

All Categories

All Forms

Any Stock Level

Prescription Only

Active Ingredient: Amoxicillin

Clear all filters

TOTAL INVENTORY VALUE

\$284,592.00

+4.2% from last month

Search Results (148 items found)

Medicine Name

SKU

Category

Stock Level

Price per Unit

Actions

Amoxicillin 500mg

Capsule • Rx

AMX-500-CP

Antibiotics

850 Units

\$0.45

View Details

+ Add to Invoice

Lisinopril 10mg

Tablet • Rx

LIS-010-TB

Cardiovascular

12 Units

\$1.12

View Details

+ Add to Invoice

Hydrocortisone 1%

Cream • OTC

HYD-001-CR

Dermatological

45 Tubes

\$8.99

View Details

+ Add to Invoice

Metformin 850mg

Tablet • Rx

MET-850-TB

Endocrinology

1,240 Units

\$0.28

View Details

+ Add to Invoice

Showing 1 to 10 of 148 entries

1

2

3

...

15

Bulk Order Recommendation

Based on recent prescription trends, it is recommended to re-order 500 units of Lisinopril 10mg to maintain optimal branch stock levels.

Review Order

Table 3-6-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Quick Search	Text	No	100	Search field to find medicines, SKUs, or invoices.
2	Category	Dropdown	No	-	Filters search results by specific medicine categories (e.g., Antibiotics, Cardiovascular).
3	Dosage Form	Dropdown	No	-	Filters results based on the medicine's form, such as Tablet, Capsule, or Cream.
4	Availability	Dropdown	No	-	Filters results based on current stock levels.
5	Clear all filters	Button	No	-	Removes all active search filters to reset the results view.
6	Medicine Name	Label	N/A	-	Displays the name, strength, and form of the medicine (e.g., Metformin 850mg Tablet).
7	SKU	Label	N/A	-	Displays the unique Stock Keeping Unit identifier for the item.
8	Stock Level	Data/Visual	N/A	-	Shows a visual progress bar and numerical count of units available at the branch.

9	Price per Unit	Label	N/A	-	Displays the current unit price for the specific medicine.
10	View Details	Button	N/A	-	Navigates the user to a detailed view of the specific medicine's information.
11	Add to Invoice	Button	N/A	-	Adds the selected medicine directly to a new or pending prescription invoice.
12	Total Inventory Value	Label	N/A	-	Displays the aggregate value of all items currently in stock for the branch.
13	Review Order	Button	N/A	-	Located in the Bulk Order Recommendation section; allows users to review suggested re-orders based on trends.
14	Export/Print	Icons	N/A	-	Allows the user to download the search results list or print the current view.

3.6.1.2 Use Case Description

Use Case ID	UC-16	Use Case Name View Medicine Information			
Author	Phạm Vũ Đình Trường	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Staff				

Description	Allow staff to search the global formulary and manage local branch inventory by filtering and viewing medicine details	
Precondition	User is successfully authenticated and logged into the system	
Trigger	Staff selects "Medicine Search" from the sidebar navigation menu	
Post-Condition	System displays matching medicine records and allows further actions (View Details, Add to Invoice)	
Main flows		
Step	Actor	Action
1	Staff	Select "Medicine Search" from the sidebar menu
2	System	Display search interface with filters for Category, Dosage Form, and Availability
3	Staff	Enter search keywords in Quick Search or select specific filters
4	System	Filter medicine database in real-time based on criteria
5	System	Display results list including Medicine Name, SKU, Stock Level, and Price per Unit
6	Staff	Click "View Details" for full info or "Add to Invoice" to start a prescription

Alternative flows

4a	No search results found	
Sub step	Actor	Action
4a.1	System	Display "No search results" message (MSG01).
4a.2	Staff	Click "Clear all filters" or modify search term.
4b	Internal System Error	
Sub step	Actor	Action
4b.1	System	Display "Error" page with message "Internal System Error, please contact with administrator".

Business Rules

#	Rule Description
BR-01	Synchronize stock levels in real-time as transactions are completed or inventory is updated.

#	Field Name	Type	Mandatory	Max Length	Description
1	Drafting ID	Label	N/A	20	Displays the temporary system-generated ID for the current draft (e.g., INV-2024-0892).
2	Search Patient / CID	Text	Yes	100	Input field to search for existing patients by name or unique Customer ID.
3	Contact Number	Text	Yes	15	Displays or allows entry of the patient's contact phone number.
4	Address / Insurance Group	Text	Yes	250	Displays or allows entry of the patient's home address and insurance provider details.
5	Medicine / SKU	Label	N/A	-	Displays the name, strength, batch, and SKU of the prescribed medicine.
6	QTY	Text	Yes	10	Input field for the number of units/tablets being dispensed for each item.
7	Unit Price	Label	N/A	-	Displays the individual price per unit for the selected medicine.
8	Item Total	Label	N/A	-	Displays the calculated total for the line item (QTY × Unit Price).

9	Add Item	Button	No	-	Opens a selection interface to add more medicines to the current invoice.
10	Remove Item	Icon	No	-	Deletes the specific medicine line item from the invoice draft.
11	Date Issued	Label	N/A	-	Displays the date the invoice is being generated (e.g., 24 Oct 2024).
12	Doctor Name	Text/Label	Yes	100	The name of the physician who issued the prescription.
13	Prescription Ref	Text/Label	Yes	50	The unique reference number for the medical prescription (e.g., RX-99012-A).
14	Payment Selection	Radio	Yes	-	Allows the user to select the payment method: Credit/Debit Card, Cash, or Insurance Direct.
15	Notes & Instructions	Text Area	No	500	Field for additional details, such as patient requests or delivery notes.
16	Subtotal	Label	N/A	-	Displays the sum of all item totals before any adjustments.
17	Insurance Coverage	Label	N/A	-	Displays the calculated discount based on the patient's insurance plan (e.g., 80% coverage).

18	Tax	Label	N/A	-	Displays the calculated tax (e.g., Exempted - Medical).
19	Payable Amount	Label	N/A	-	Displays the final total amount due from the customer.
20	Save Draft	Button	No	-	Saves the current invoice progress without finalizing the transaction.
21	Generate Invoice	Button	Yes	-	Finalizes the transaction, records the sale, and triggers the deduction of quantities from inventory.

3.6.2.2 Use Case Description

Use Case ID	UC-15	Use Case Name		Invoice Generation	
Author	Phạm Vũ Đình Trường	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Staff				
Description	Create sales invoices for customers based on prescriptions				
Precondition	Staff is successfully authenticated and logged into the system				
Trigger	Staff clicks "+ New Prescription" or "Add to Invoice" from medicine search				
Post-Condition	Invoice is recorded, and quantities are deducted from inventory				
Main flows					
Step	Actor		Action		
1	Staff		Request to create a new prescription invoice by clicking "+ New Prescription" or "Add to Invoice"		
2	System		Display "Prescription Invoice" screen with a unique Drafting ID		
3	Staff		Provide Patient / CID to search for and identify the customer		

4	System	Retrieve and display patient contact information, address, and insurance group
5	Staff	Add prescribed medicines to the list and specify the quantity (QTY) for each item
6	System	Calculate line item totals, subtotal, insurance coverage, tax, and final payable amount
7	Staff	Select payment method and enter doctor name and prescription reference
8	Staff	Click "Generate Invoice" button to finalize the transaction
9	System	Record the transaction, deduct quantities from branch inventory, and synchronize data

Alternative flows

4a	Insufficient Stock	
Sub step	Actor	Action
4a.1	System	Notify that requested quantity exceeds available stock level.
4a.2	Staff	Adjust quantity or select a different medicine.
7a	Missing Required Information	
Sub step	Actor	Action
7a.1	System	Highlight empty mandatory fields (e.g., Doctor Name, Prescription Ref).
7a.2	Staff	Provide missing information and return to Step 7 of Main Flow.

Business Rules

#	Rule Description
BR-01	Automatically deduct quantities from branch inventory upon invoice generation.
BR-02	Protect PII and financial information using 256-bit encryption during network transmission.

BR-03	Ensure all items in a single order are paid for using the same payment method.
BR-04	Calculate total price as the sum of items times quantity, minus insurance coverage, plus applicable tax.
BR-05	Restrict "Generate Invoice" functionality to staff members assigned to the specific branch.

3.6.3 <<UC18-View Generated Invoices>>

3.6.3.1 Screen Mock-up

The screenshot shows the 'Invoices' screen in the Stratos Enterprise system for Pharmacy Branch 042. The interface includes a sidebar with navigation links: Dashboard, Medicine Search, Invoices (selected), and Profile. The main content area features a search bar at the top, followed by a summary section with 'Daily Revenue: \$12,450.00' and 'Pending: 24 items'. Below this is a table of invoices with columns: Invoice ID, Date Issued, Customer Name, Total Amount, Payment Status, and Actions. The table lists four invoices, with the first two marked as 'Paid' and the last two as 'Pending'. At the bottom, there are sections for 'Quarterly Audit Ready' status and 'Quick Help' links.

Invoice ID	Date Issued	Customer Name	Total Amount	Payment Status	Actions
#INV-8902	Oct 24, 2023	Alexander Pierce	\$450.00	Paid	Print, Details
#INV-8903	Oct 25, 2023	Sarah J. Miller	\$1,280.50	Pending	Print, Process Payment
#INV-8904	Oct 25, 2023	Dominic Torretto	\$89.99	Paid	Print, Details
#INV-8905	Oct 26, 2023	Elena Rodriguez	\$312.00	Pending	Print, Process Payment

Table 3-6-3: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Search Bar	Text	No	100	Allows searching for specific records by Invoice ID or customer names.

2	Status Filter	Dropdown	No	-	Filters the invoice list based on payment status (e.g., All Statuses, Paid, Pending).
3	Date Range	Date Picker	No	-	Filters the displayed invoices to a specific issuance date or range.
4	Apply Filters	Button	No	-	Triggers the system to update the list based on search and filter parameters.
5	Daily Revenue	Label	N/A	-	Displays a summary of the total revenue generated for the current day.
6	Pending Items	Label	N/A	-	Displays a summary count of all invoices currently in "Pending" status.
7	Download/Export	Icon	No	-	Exports the currently filtered list of invoices to an external file.
8	Print (List)	Icon	No	-	Opens a print dialog for the current table view.
9	Invoice ID	Label	N/A	20	Displays the unique identifier for the transaction (e.g., #INV-8902).
10	Date Issued	Label	N/A	-	Displays the date the invoice was created.

11	Customer Name	Label	N/A	100	Displays the name of the patient associated with the transaction.
12	Total Amount	Label	N/A	-	Displays the final payable amount for the invoice.
13	Payment Status	Badge	N/A	-	Visual indicator of whether the invoice is "Paid" (green) or "Pending" (red).
14	Print (Action)	Button	No	-	Triggers the document generation and printing for a specific invoice.
15	Details	Button	No	-	Navigates the user to a comprehensive view of the specific invoice details.
16	Process Payment	Button	No	-	Active for pending items; redirects to the Payment Processing interface to complete the sale.
17	Pagination	Control	No	-	Enables navigation through multiple pages of invoice records.

3.6.3.2 Use Case Description

Use Case ID	UC-18	Use Case Name		View Generated Invoice	
Author	Phạm Vũ Đình Trường	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Staff				
Description	Allows staff to view previously generated invoices and manage payment statuses.				
Precondition	User is successfully authenticated and authorized for the branch.				

Trigger	Staff selects the "Invoices" tab from the sidebar menu.	
Post-Condition	System displays relevant transaction history for the branch.	
Main flows		
Step	Actor	Action
1	Staff	Request to view invoices by selecting the "Invoices" menu.
2	System	Retrieve and display the invoice list for the assigned branch.
3	Staff	Enter search terms or select status and date filters.
4	Staff	Click "Apply Filters".
5	System	Update the list to show matching invoice records.
6	Staff	Select "Details" to view items or "Process Payment" for pending items.

Alternative flows

4a	No search results found	
Sub step	Actor	Action
4a.1	System	Display message MSG01: "There is not any search result".
4a.2	Staff	Modify filters or search terms and return to Step 4 of Main Flow.
5a	Internal System Error	
Sub step	Actor	Action
5a.1	System	Display error page with message "Internal System Error, please contact with administrator".

Business Rules

#	Rule Description
BR-01	Encrypt network transmissions involving financial information using 256-bit encryption.
BR-02	Restrict staff access to invoices generated within their specifically assigned branch.

BR-03	Synchronize "Payment Status" in real-time with the Payment Processing module.
BR-04	Prevent modification of invoice items or amounts if the status is "Paid".

3.6.4 <<UC17-Payment Processing>>

3.6.4.1 Screen Mock-up

Stratos Health

Checkout
Invoice #INV-2024-042

PATIENT DETAILS
Alexander Vance
Medical ID: PH-9902-X • DOB: 12 May 1985

Prescribed Items
Qty: 04

	Amoxicillin 500mg 30 Capsules • Prescription 88412	\$24.50
	Lisinopril 10mg 90 Tablets • Prescription 88413	\$18.25
	B12 Multi-Vitamin Liquid 150ml Oral Solution	\$32.10

Branch Totals

Subtotal	\$74.85
Tax (8%)	\$5.99

TOTAL PAYABLE
\$80.84

Payment Method

Credit / Debit Card
Visa, Mastercard, Amex
☐

Cash Payment
Physical Currency at Terminal
☐

Insurance Claim
Policy: STRT-0098412-V
☐

Confirm Payment

Secure 256-bit SSL encrypted connection

Table 3-6-4: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Invoice Number	Label	N/A	-	Displays the unique invoice ID (e.g., #INV-2024-042).

2	Patient Name	Label	N/A	-	Displays the full name of the patient.
3	Medical ID	Label	N/A	-	Displays the unique medical identification code for the patient.
4	Date of Birth	Label	N/A	-	Displays the patient's birth date.
5	Prescribed Items List	Table/Label	N/A	-	Lists medicines, quantities, and individual prices.
6	Subtotal	Label	N/A	-	Sum of all item prices before taxes.
7	Tax (%)	Label	N/A	-	Calculated tax amount (e.g., 8%) applied to the subtotal.
8	Total Payable	Label	N/A	-	The final total amount due from the customer.
9	Payment Method	Radio	Yes	-	Options: Credit/Debit Card, Cash, or Insurance Claim.
10	Confirm Payment	Button	Yes	-	Initiates the transaction with the selected payment method.

3.6.4.2 Use Case Description

Use Case ID	UC-17	Use Case Name Payment Processing			
Author	Phạm Vũ Đình Trường	Version	1.0	Date	02/06/2026
Actor	Primary Actor: Staff, Secondary Actor: Payment Gateway				

Description	Allows staff to process and record the final payment for a generated invoice, integrating with external services to verify status	
Precondition	An invoice has been generated and is in "Pending" status	
Trigger	Staff clicks "Process Payment" from the View Generated Invoices screen	
Post-Condition	Invoice status is updated to "Paid" and the transaction confirmation is recorded	
Main flows		
Step	Actor	Action
1	Staff	Request to process payment for a specific pending invoice.
2	System	Display Checkout screen with patient details, itemized list, and totals
3	Staff	Verify the final payable amount with the customer
4	Staff	Select the customer's preferred Payment Method
5	Staff	Click "Confirm Payment"
6	System	Connect to external Payment Gateway to verify the electronic transaction
7	System	Receive payment confirmation, update invoice status to "Paid", and display success message

Alternative flows

6a	Payment Gateway Connection Error	
Sub step	Actor	Action
6a.1	System	Notify connection failure and prompt to try again or use a different payment method.
6a.2	Staff	Re-select payment method or retry and return to Step 5 of Main Flow.
6b	Payment Declined	
Sub step	Actor	Action
6b.1	System	Notify that transaction was declined by bank or insurance provider. clicks "Cancel" or navigates away before saving
6b.2	Staff	Request an alternative payment method from customer. Terminates the edit process without saving changes and returns to the view-only profile state

Business Rules

#	Rule Description
BR-01	Encrypt network transmissions involving financial information using 256-bit SSL encryption.
BR-02	Require a successful return code from the Payment Gateway before marking a transaction as completed.
BR-03	Ensure all items in a single order are paid for using the same payment method.
BR-04	Prevent further modification of invoice details or quantities once the status is updated to "Paid".

3.7. Inventory Manager

3.7.1.<<UC31-Track Batch & Lot Number>>

3.7.1.1 Screen Mock-up

LogisticsPro
Warehouse 01

Search POs, Items...

Receive Goods

Inventory Manager Admin

Receipt Details

VENDOR: Acme Electronics Ltd. PURCHASE ORDER NUMBER: PO-2023-8842 DELIVERY DATE: 10/25/2023 RECEIVING LOCATION: Dock 2 (Electronics)

Incoming Items Partially Received [+ Add Item](#)

PRODUCT CODE	PRODUCT NAME	ORDERED QTY	RECEIVED QTY	BATCH NUMBER	EXPIRY DATE
☐ ELC-992-A	Capacitor 100uF 50V	5000	5000	B-4492-X	12/31/2028
☒ ELC-104-B	Resistor 10k Ohm 1/4W	10000	8500	B-4493-Y	05/15/2030
☐ MCU-ESP32-WROVER	ESP32 WROVER Module	200	0	Scan Batch...	mm/dd/yyyy
☐ PWR-5V-2A	5V 2A Power Supply	500	0	Scan Batch...	mm/dd/yyyy

Cancel Save Draft Submit Receipt

Profile Logout

Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Vendor	Drop-down	Yes	-	Select supplier delivering goods.
2	Purchase Order	Drop-down	Yes	-	Select associated purchase order.
3	Delivery Note Number	Text	Yes	50	Vendor delivery reference number.
4	Received Date	Date Picker	Yes	-	Date goods are received.

5	Product List	Table	Yes	-	List of received products.
6	Received Quantity	Number	Yes	-	Quantity received for each item.
7	Remarks	Text Area	No	500	Additional receiving notes.
8	Save	Button	-	-	Saves goods receipt information.
9	Cancel	Button	-	-	Cancels operation and returns.

Table 3-1: Screen Definition

3.7.2.2 Use Case Receive Goods from Vendor

Use Case ID	UC07	Use Case Name		Receive Goods from Vendor	
Author	HungNH	Version	1.0	Date	02/06/2026
Actor	Inventory Manager				
Description	Inventory Staff records goods received from suppliers into warehouse inventory.				
Precondition	Authorized user is logged in. Purchase order exists.				
Trigger	Inventory Staff receives goods from supplier.				
Post-Condition	Inventory quantity is updated and receipt record is created.				
Main flows					
Step	Actor		Action		

1	Inventory Staff	Requests goods receiving function.
2	System	Displays receiving form.
3	Inventory Staff	Enters receipt information.
4	System	Validates received data.
5	System	Creates receiving record and updates stock.

Alternative flows

4a	Invalid Data	
Step	Actor	Action
4a.1	System	Displays validation errors.
4a.2	Inventory Staff	Corrects information.
4b	Quantity Exceeds PO	
Step	Actor	Action
4b.1	System	Warns quantity exceeds approved PO.
4b.2	Founder	Provides a different, unique email address (Returns to step 4).

Business Rules

#	Rule Description
BR-01	Received quantity cannot exceed PO quantity without approval.

BR-02	All received items must be linked to a valid vendor.
BR-03	Receipt transactions must be auditable.

3.7.3.<<UC08-QC Inspection>>

3.7.3.1 Screen Mock-up

LogisticsPro
Warehouse 01

Inventory Management

QC Inspection
Review and process received inventory for quality control.

PENDING INSPECTIONS: 15
PASSED: 142
FAILED: 3
UNDER REVIEW: 2

Recent Receipts

RECEIPT ID	PRODUCT NAME	BATCH NUMBER	QUANTITY	INSPECTION STATUS	INSPECTOR	ACTION
RCPT-8092	Surgical Masks (N95)	BCH-2023X-A1	5,000	PENDING	-	INSPECT
RCPT-8091	Nitrile Gloves (Large)	BCH-2023X-B2	10,000	PASSED	J. Doe	VIEW
RCPT-8090	Saline Solution (500ml)	BCH-2023X-C3	2,500	FAILED	A. Smith	VIEW

Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Receipt Number	Drop-down	Yes	-	Select goods receipt for inspection.

2	Product Information	Table	-	-	Displays products awaiting inspection.
3	Inspection Result	Drop-down	Yes	-	Passed, Rejected, or Pending Review.
4	Defect Description	Text Area	No	500	Details of detected issues.
5	Inspection Date	Date Picker	Yes	-	Date inspection is performed.
6	Inspector Name	Text	Auto	100	Logged-in inspector.
7	Save Inspection	Button	-	-	Saves inspection result.
8	Cancel	Button	-	-	Cancels inspection process.

3.7.3.2 Use Case QC Inspection

Use Case ID	UC08	Use Case Name	QC Inspection		
Author	HungNH	Version	1.0	Date	02/06/2026
Actor	Inventory Manager				
Description	QC Staff inspects received goods before releasing them into inventory.				

Precondition	Goods receipt exists.	
Trigger	Goods require quality inspection.	
Post-Condition	Goods are marked as Passed, Rejected, or Pending Review.	
Main flows		
Step	Actor	Action
1	Inventory Manager	Requests to perform quality inspection on received goods.
2	System	Displays the list of received goods awaiting inspection.
3	Inventory Manager	Selects a goods receipt and enters inspection results.
4	System	Validates the inspection information.
5	System	Updates the inspection status and records the inspection result.
6	System	Marks goods as Passed, Rejected, or Pending Review.

Alternative flows

4a	Invalid Inspection Data	
Step	Actor	Action
4a.1	System	Displays validation errors and requests correction.
4a.2	Inventory Manager	Corrects the inspection information and resubmits (Returns to Step 4).
6a	Goods Rejected	

Step	Actor	Action
6a.1	System	Marks goods as Rejected.
6a.2	System	Moves the goods to quarantine/rejected inventory.

*a	Cancellation	
Step	Actor	Action
*a.1	Inventory Manager	Requests to cancel the inspection process.
*a.2	System	Terminates the operation and returns to the previous screen.

Business Rules

#	Rule Description
BR-01	Goods must pass QC inspection before being available for inventory issuance.
BR-02	Rejected goods must be isolated from available inventory.
BR-03	Inspection results must be recorded and auditable.
BR-04	Only authorized Inventory Managers or QC Staff may perform inspections.

3.7.4.<<UC11-Issue Stock to Stores>>

3.7.4.1 Screen Mock-up

LogisticsPro
Warehouse 01

Dashboard
Receive Goods
QC Inspection
Issue Stock
Internal Transfers
Stocktake
Safety Stock

Issue Stock

Search requests...

Inventory Manager Admin

PENDING REQUESTS: 7
APPROVED REQUESTS: 24
ISSUED TODAY: 156 units
TOTAL QTY ISSUED: 4.2k

Issue Requests

REQUEST ID	STORE NAME	PRODUCT	REQ QTY	APPR QTY	STATUS	ACTIONS
REQ-9012	Downtown Flagship	SKU-A12 Ergo Chair Pro	50	-	Pending	[Icon]
REQ-9008	Northside Mall	SKU-B44 Standing Desk	15	15	Approved	[Icon]
REQ-8995	Westend Plaza	SKU-C01 Monitor Arm Dual	100	95	Issued	[Icon]
REQ-9015	Eastside Outlet	SKU-D22 Filing Cabinet	12	-	Pending	[Icon]

Showing 1 to 4 of 31 entries

Profile
Logout

Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Request Number	Drop-down	Yes	-	Select approved stock request.
2	Store/Department	Drop-down	Yes	-	Destination store or department.
3	Product List	Table	Yes	-	Products requested for issue.
4	Issue Quantity	Number	Yes	-	Quantity issued.

5	Issue Date	Date Picker	Yes	-	Date stock is issued.
6	Remarks	Text Area	No	500	Additional notes.
7	Confirm Issue	Button	-	-	Completes stock issue transaction.
8	Cancel	Button	-	-	Cancels operation.

3.7.4.2 Use Case Issue Stock to Stores

Use Case ID	UC11	Use Case Name		Issue Stock to Stores	
Author	HungNH	Version	1.0	Author	HungNH
Date	02/06/2026				
Actor	Inventory Manager				
Description	Inventory Manager issues inventory items from the warehouse to stores or departments based on approved requests.				
Precondition	Stock request exists and sufficient inventory is available.				
Trigger	Store submits an approved stock request.				
Main flows					
Step	Actor		Action		
1	Inventory Manager		Requests to issue stock to a store.		

2	System	Displays pending stock requests.
3	Inventory Manager	Selects a request and specifies the quantity to issue.
4	System	Validates stock availability and request information.
5	System	Records the stock issue transaction.
6	System	Updates inventory quantities and confirms successful issuance.

Alternative flows

4a	Insufficient Stock	
Step	Actor	Action
4a.1	System	Notifies that available stock is insufficient.
4a.2	Inventory Manager	Adjusts the issue quantity or cancels the transaction.
4b	Invalid Request	
Step	Actor	Action
4b.1	System	Displays an error indicating invalid or closed request.
4b.2	Inventory Manager	Selects another valid request (Returns to Step 3).

Business Rules

#	Rule Description
BR-01	Issued quantity must not exceed available stock.
BR-02	Every stock issue transaction must reference a valid request.
BR-03	All stock issue activities must be auditable.

3.7.5.<<UC12-Approve Internal Transfers>>

3.7.5.1 Screen Mock-up

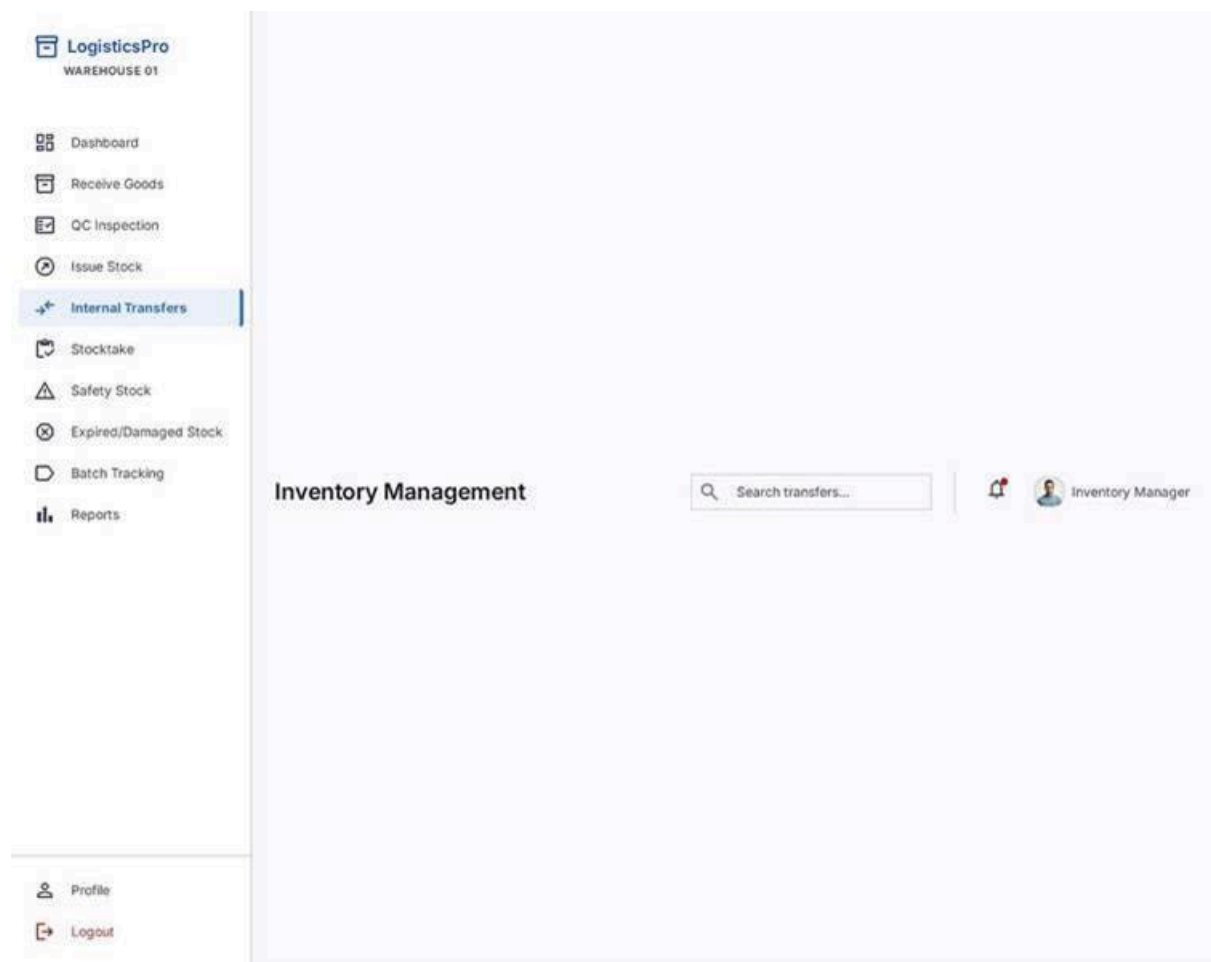


Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description

1	Transfer Request No.	Text	-	30	Unique transfer request identifier.
2	Source Warehouse	Text	-	100	Warehouse transferring stock.
3	Destination Warehouse	Text	-	100	Warehouse receiving stock.
4	Product List	Table	-	-	Products included in transfer.
5	Requested Quantity	Number	-	-	Quantity requested for transfer.
6	Approval Status	Drop-down	Yes	-	Approve or Reject.
7	Rejection Reason	Text Area	No	500	Required when rejecting request.
8	Approve Transfer	Button	-	-	Approves transfer request.
9	Reject Transfer	Button	-	-	Rejects transfer request.

3.7.5.2 Use Case Approve Internal Transfers

Use Case ID	UC12	Use Case Name		Approve Internal Transfers	
Author	HungNH	Version	1.0	Author	HungNH
Date	02/06/2026				
Actor	Inventory Manager				

Description	Inventory Manager reviews and approves stock transfer requests between warehouses or storage locations.	
Precondition	Transfer request exists.	
Trigger	A transfer request is submitted for approval.	
Main flows		
Step	Actor	Action
1	Inventory Manager	Opens the transfer approval screen.
2	System	Displays pending transfer requests.
3	Inventory Manager	Reviews transfer details.
4	Inventory Manager	Approves the transfer request.
5	System	Updates transfer status to Approved.
6	System	Creates transfer records and notifies related parties.

Alternative flows

4a	Transfer Rejected	
Step	Actor	Action
4a.1	Inventory Manager	Rejects the transfer request.
4a.2	System	Updates the status to Rejected and records the reason.
4b	Insufficient Source Stock	

Step	Actor	Action
4b.1	System	Detects insufficient stock at source location.
4b.2	System	Prevents approval and displays an error message.

Business Rules

#	Rule Description
BR-01	Only authorized Inventory Managers may approve transfers.
BR-02	Transfer quantity must not exceed available stock.
BR-03	All approvals and rejections must be logged.

3.7.6.<<UC19-Set Safety Stock Levels>>

3.7.6.1 Screen Mock-up

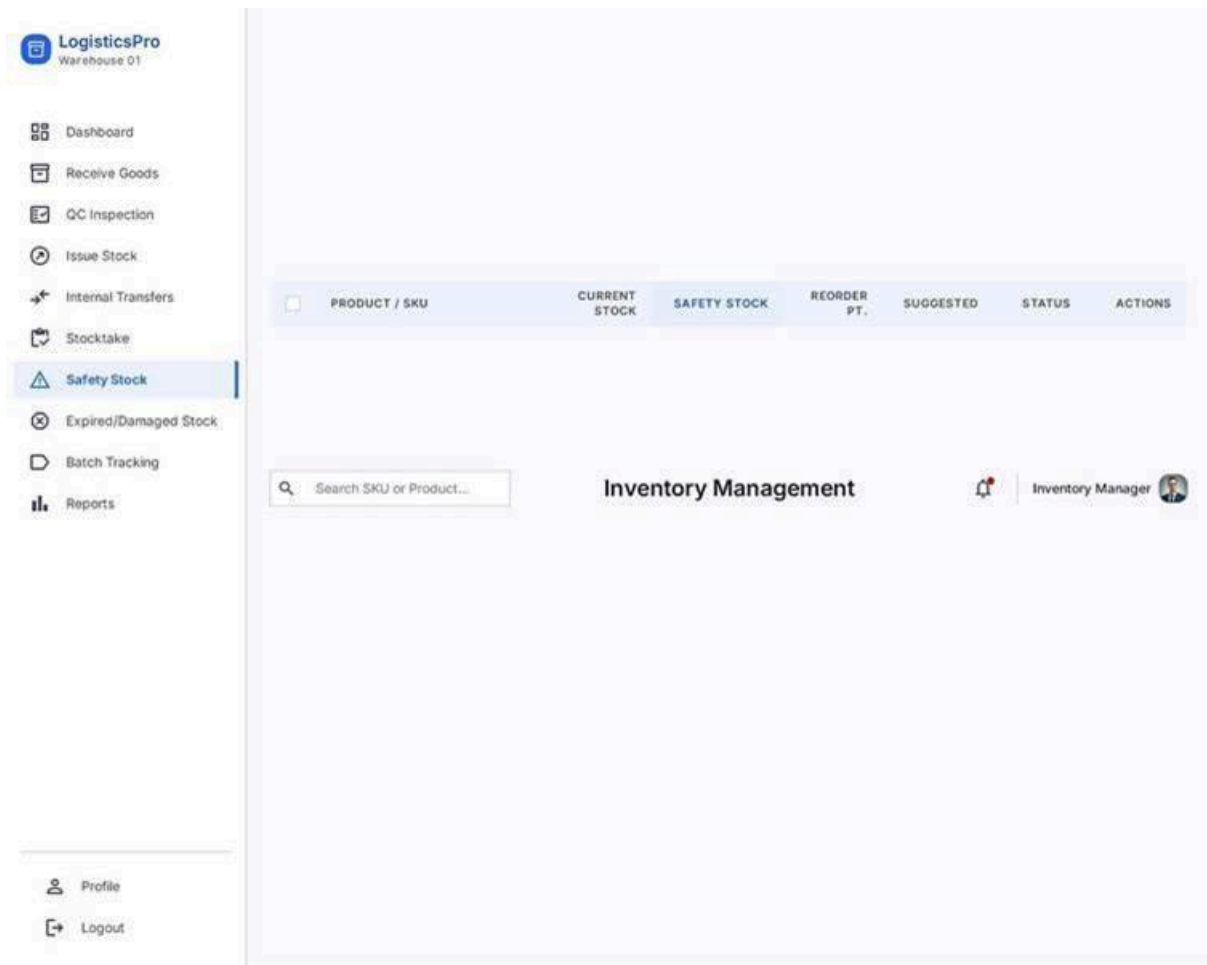


Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Product	Drop-down	Yes	-	Select inventory product.
2	Current Stock	Number	-	-	Current inventory quantity.
3	Safety Stock Level	Number	Yes	-	Minimum acceptable stock quantity.
4	Alert Enabled	Checkbox	Yes	-	Enables low stock notification.

5	Save Settings	Button	-	-	Saves safety stock configuration.
6	Cancel	Button	-	-	Cancels changes.

3.7.6.2 Use Case Set Safety Stock Levels

Use Case ID	UC19	Use Case Name		Set Safety Stock Levels	
Author	HungNH	Version	1.0	Author	HungNH
Date	02/06/2026				
Actor	Inventory Manager				
Description	Inventory Manager defines minimum inventory thresholds to prevent stock shortages.				
Precondition	Product records exist.				
Trigger	Inventory policy requires update.				
Main flows					
Step	Actor		Action		
1	Inventory Manager		Opens safety stock configuration screen.		
2	System		Displays inventory items.		
3	Inventory Manager		Selects a product and enters safety stock level.		
4	System		Validates the entered value.		

5	System	Saves the safety stock level.
6	System	Activates monitoring for stock alerts.

Alternative flows

4a	Invalid Safety Stock Value	
Step	Actor	Action
4a.1	System	Displays validation error.
4a.2	Inventory Manager	Enters a valid safety stock value.

Business Rules

#	Rule Description
BR-01	Safety stock level must be greater than or equal to zero.
BR-02	The system must generate alerts when stock falls below the threshold.
BR-03	Only authorized users may modify safety stock levels.

3.7.7.<<UC30-Identify Expired/Damaged Stock>>

3.7.7.1 Screen Mock-up

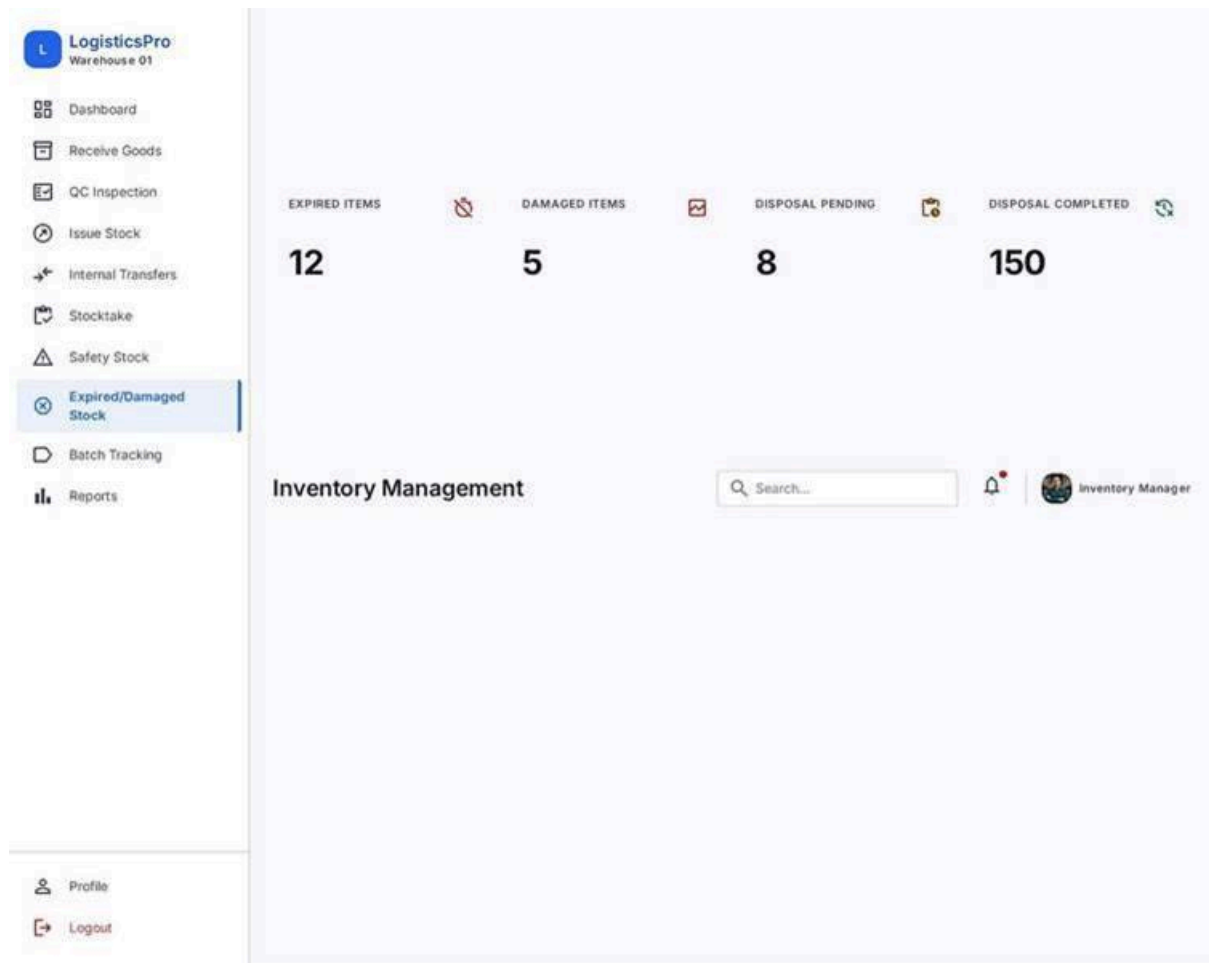


Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Product Search	Text	No	100	Search inventory items.
2	Product List	Table	-	-	Displays inventory items.
3	Expiry Date	Date	-	-	Product expiration date.

4	Stock Condition	Drop-down	Yes	-	Normal, Expired, or Damaged.
5	Damage Description	Text Area	No	500	Description of damage.
6	Update Status	Button	-	-	Updates stock condition.
7	Cancel	Button	-	-	Cancels operation.

3.7.7.2 Use Case Identify Expired/Damaged Stock

Use Case ID	UC30	Use Case Name	Identify Expired/Damaged Stock		
Author	HungNH	Version	1.0	Date	02/06/2026
Actor	Inventory Manager				
Description	Inventory Manager identifies inventory items that are expired or damaged.				

Precondition	Inventory records contain expiry date or condition information.	
Trigger	Routine inspection or stock review is performed.	
Post-Condition	Affected inventory items are marked and isolated.	
Main flows		
Step	Actor	Action
1	Inventory Manager	Opens inventory inspection screen.
2	System	Displays inventory items and expiry information.
3	Inventory Manager	Reviews items and selects expired/damaged stock.
4	System	Validates selected items.
5	System	Updates stock status to Expired or Damaged.
6	System	Removes affected stock from available inventory.

Alternative flows

3a	No Expired/Damaged Items Found
-----------	--------------------------------

Step	Actor	Action
3a.1	Inventory Manager	Completes inspection.
3a.2	System	Records inspection result with no issues found.

Business Rules

#	Rule Description
BR-01	Expired stock cannot be issued or transferred.
BR-02	Damaged stock must be isolated from usable inventory.
BR-03	All identified items must be traceable.

4. Non-Functional Requirements

4.1 External Interfaces

- **EI-1: External Database Integration:** The system **MUST** integrate with the National Prescription Database (or regional regulatory equivalent) via RESTful API, ensuring data synchronization within 2 minutes of a prescription status update.
 - **Justification:** Pharmacy chains must comply with real-time legal regulations regarding controlled substances to avoid dispensing restricted drugs improperly or illegally.
- **EI-2: Hardware Interoperability:** The point-of-sale (POS) subsystem **MUST** support standard receipt thermal printers using standard ESC/POS commands, maintaining a print execution delay of less than 1.5 seconds per receipt command.
 - **Justification:** Delays in hardware communication at the checkout counter directly cause customer queues to build up during peak retail hours.

4.2 Quality Attributes

4.2.1 Usability

- **U-1: Transaction Efficiency:** The POS and prescription checkout UI **MUST** allow a trained pharmacist to complete a standard prescription verification and sale within a maximum of 5 mouse clicks and under 25 seconds.
 - **Justification:** High foot traffic in pharmacies requires highly optimized workflows. Minimizing clicks reduces physical fatigue and accelerates the queue processing rate.
- **U-2: Error Rate on Data Entry:** The system **MUST** utilize strict input masking and auto-completion for drug names, ensuring that input errors during manual prescription entry occur in fewer than 1% of total daily transactions.
 - **Justification:** Medication dispensing errors can be fatal. Forcing strict format inputs mitigates human typing errors common in long shifts.

4.2.2 Reliability

- **R-1: System Availability:** The system **MUST** guarantee an overall availability (uptime) of 99.99% between the operating hours of 6:00 AM and 11:00 PM local time, excluding pre-scheduled maintenance windows.
 - **Justification:** Pharmacies provide critical health services. A system outage directly halts the ability to dispense life-saving medications and process critical insurance claims.
- **R-2: Recovery Time Objective (RTO):** In the event of a critical server failure, the system's RTO **MUST NOT** exceed 15 minutes to restore core POS and dispensing functionalities.
 - **Justification:** 15 minutes is the maximum acceptable threshold before a retail pharmacy branch begins losing significant revenue, and patient trust is severely damaged due to inability to process orders.

4.2.3 Performance

- **P-1: Query Response Time:** The system **MUST** execute and return results for 95% of standard inventory search queries (e.g., checking stock for a specific drug at a local branch) within 1.2 seconds.
 - **Justification:** Pharmacists and technicians require immediate answers regarding drug availability when a patient is waiting at the counter or calling on the phone.
- **P-2: System Throughput:** The central application server **MUST** support a minimum processing throughput of 500 Transactions Per Second (TPS) during the daily peak hours of 4:00 PM to 7:00 PM.
 - **Justification:** Across a multi-store pharmacy chain, the aggregate number of simultaneous checkouts, inventory deductions, and prescription verifications peaks heavily after standard work hours.

4.2.4 Security

- **S-1: Data Encryption at Rest:** All Patient Health Information (PHI), including prescription histories and personal identities, **MUST** be encrypted at rest using the AES-256 encryption standard.
 - **Justification:** Pharmacy chains are legally bound by strict health data protection regulations (e.g., HIPAA) which mandate cryptographic protection for stored medical records to prevent data breaches.
- **S-2: Session Management:** User sessions for both the administrative and pharmacist portals **MUST** automatically terminate after 10 minutes of inactivity, requiring full re-authentication (Password + 2FA) to resume.
 - **Justification:** Shared computer terminals in a busy retail pharmacy environment pose a high security risk if a logged-in session is left unattended by a staff member.

4.2.5 Scalability

- **SC-1: Data Storage Growth:** The database architecture **MUST** support an ongoing annual data growth of 2 Terabytes without exceeding the 1.2-second response time metric defined in P-1.
 - **Justification:** A growing pharmacy chain accumulates massive volumes of transaction logs, multi-year prescription histories, and regulatory audit trails daily.
- **SC-2: Concurrent User Capacity:** The system **MUST** sustain concurrent usage by up to 3,000 active staff members without exceeding an average CPU utilization of 75% on the application servers.
 - **Justification:** The infrastructure must support the business's expansion plan (adding more branches and terminals) without requiring immediate and costly hardware overhauls.

4.2.6 Maintainability

- **M-1: Code Complexity Limit:** The cyclomatic complexity for all newly committed backend functions dealing with drug interactions **MUST NOT** exceed 10, as measured by standard static analysis tools (e.g., SonarQube).
 - **Justification:** Low complexity ensures that the medical logic can be rigorously unit-tested and safely updated by new developers when medical guidelines change.
- **M-2: Comprehensive Audit Logging:** The system **MUST** automatically log all drug dispensing actions, user logins, and inventory adjustments, retaining these logs in a structured, queryable database for a minimum of 5 years.
 - **Justification:** Regulatory and compliance audits in the pharmaceutical industry often look back several years; the IT system must facilitate fast retrieval of historical dispensing events.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Prescription medications must only be dispensed upon validation of an active, non-expired prescription issued by a registered medical practitioner.
BR-02	The sale of restricted over-the-counter (OTC) substances must strictly require the extraction and recording of a valid government-issued photo identification.
BR-03	Expired medications, or medications within 7 days of their expiration date, cannot be sold and must be automatically designated to the quarantine inventory.
BR-04	A primary pharmacist with an active professional license must digitally authorize the dispensing of any highly restricted or specially controlled medication.
BR-05	Any single retail transaction exceeding 20,000,000 VND must be flagged for secondary review and authorized by a shift manager before final payment processing.
BR-06	Discounts applied to retail pharmaceutical products must not reduce the final sale price below the calculated wholesale acquisition cost (WAC) without overriding approval from a regional manager.
BR-07	A patient's health insurance claim must be fully adjudicated or formally rejected by the third-party provider before the final out-of-pocket calculation is applied to the transaction.
BR-08	Inventory reconciliation for controlled substances must be conducted and logged every 24 hours to maintain strict compliance with health regulatory frameworks.

5.2 Common Requirements

- **CR-01:** All monetary values must be standardized to Vietnamese Dong (VND), utilizing a comma as the thousands separator and formatted without decimal values, appended with the currency symbol (e.g., 1,500,000 đ).

- **CR-02:** System timestamps for audit logs must be recorded in Coordinated Universal Time (UTC), but displayed to end-users in the standard Indochina Time (ICT) timezone using the format DD/MM/YYYY HH:MM (e.g., 14/10/2026 15:30).
- **CR-03:** All data tables displaying records (e.g., patient lists, inventory logs, prescription history) must support pagination with configurable page limits of 20, 50, and 100 items per page.
- **CR-04:** Any modification to inventory quantities, pricing tiers, or patient medical profiles must generate an immutable background audit trail detailing the User ID, action performed, original value, new value, and precise timestamp.
- **CR-05:** User authentication passwords must be at least 12 characters in length, containing at least one uppercase letter, one lowercase letter, one numeric digit, and one special character.

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG-001	Toast message	Upon successful validation and saving of a new patient profile.	<i>Patient profile successfully created and saved.</i>
2	MSG-002	In red, under the text box	When a user attempts to enter a non-numerical or invalid drug registration code.	<i>Please enter a valid product registration code.</i>
3	MSG-003	Pop-up	When the system detects a potential severe drug-drug interaction during prescription entry.	<i>Warning: Potential severe interaction detected with the patient's existing medication. Please review clinical pharmacology guidelines before proceeding.</i>
4	MSG-004	In line	When attempting to process a prescription for an item currently with zero stock.	<i>The selected medication is currently out of stock. Please initiate an inter-store transfer or restock request.</i>

5	MSG-005	Pop-up	When the third-party health insurance adjudication service is unresponsive or times out.	<i>Unable to connect to the insurance provider at this moment. Please verify network connectivity or process as a cash payment temporarily.</i>
6	MSG-006	In red, under the text box	When a user tries to apply a discount percentage that drops the price below the wholesale cost.	<i>Discount cannot exceed the authorized limit. Manager override required.</i>
7	MSG-007	Pop-up	When a transaction total exceeds the 20,000,000 VND limit for a standard clerk.	<i>Transaction amount exceeds the standard threshold. Please request manager approval to proceed.</i>
8	MSG-008	Toast message	When an inter-store inventory transfer request is successfully sent to another branch.	<i>Inventory transfer request successfully dispatched to the destination branch.</i>
9	MSG-009	Pop-up	When attempting to mark a batch of medication as expired or damaged.	<i>Are you sure you want to move this batch to the quarantine inventory? This action will require reconciliation approval.</i>
10	MSG-010	In line	When entering an invalid prescribing physician's credential number.	<i>The provided physician identifier could not be verified. Please check the credentials and try again.</i>
11	MSG-011	Toast message	When the daily inventory reconciliation report is successfully submitted.	<i>Daily reconciliation report securely archived.</i>

5.4 Other Requirements...

